

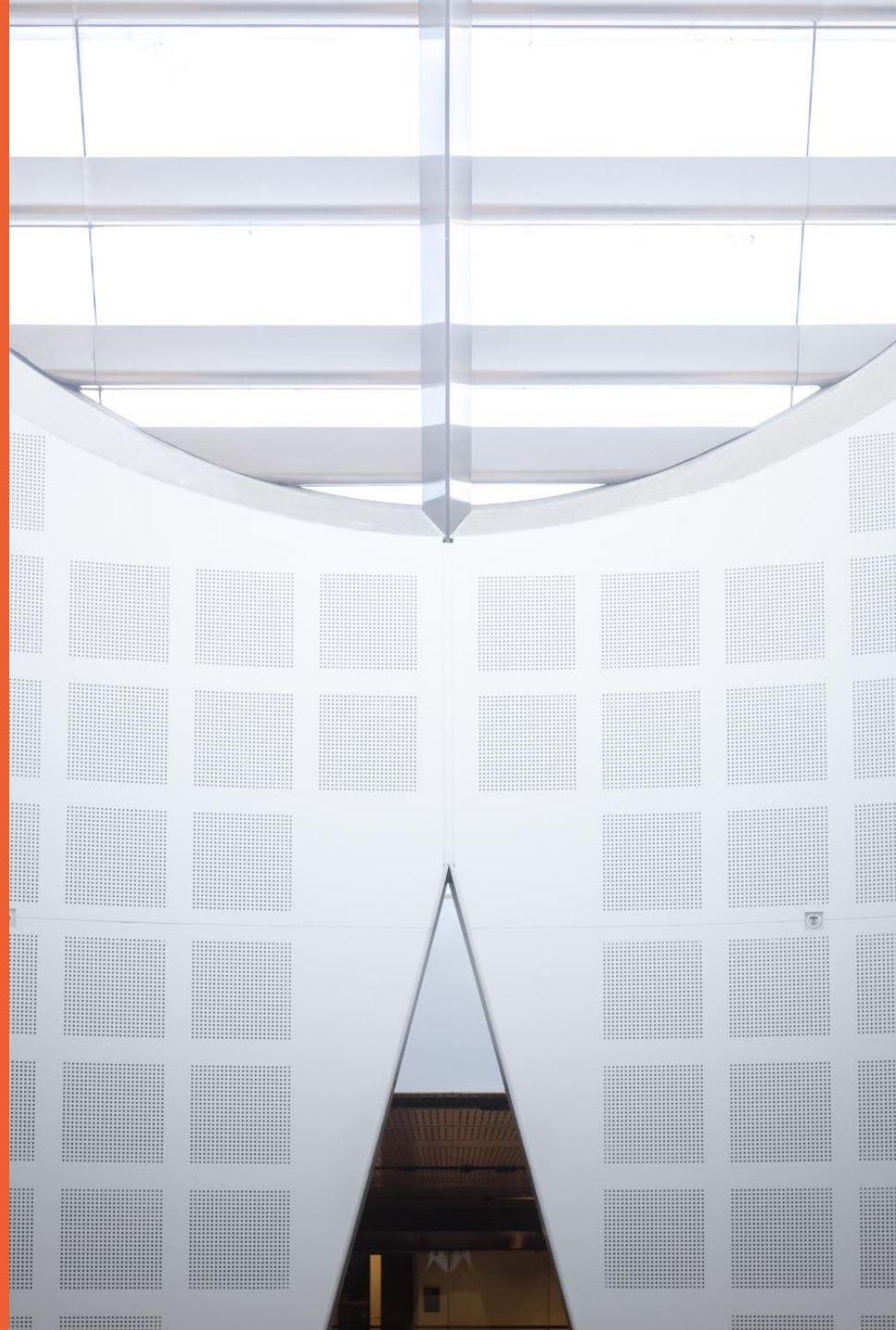
Multi-modal Foundation Model

Yunke Wang

School of Computer Science



THE UNIVERSITY OF
SYDNEY

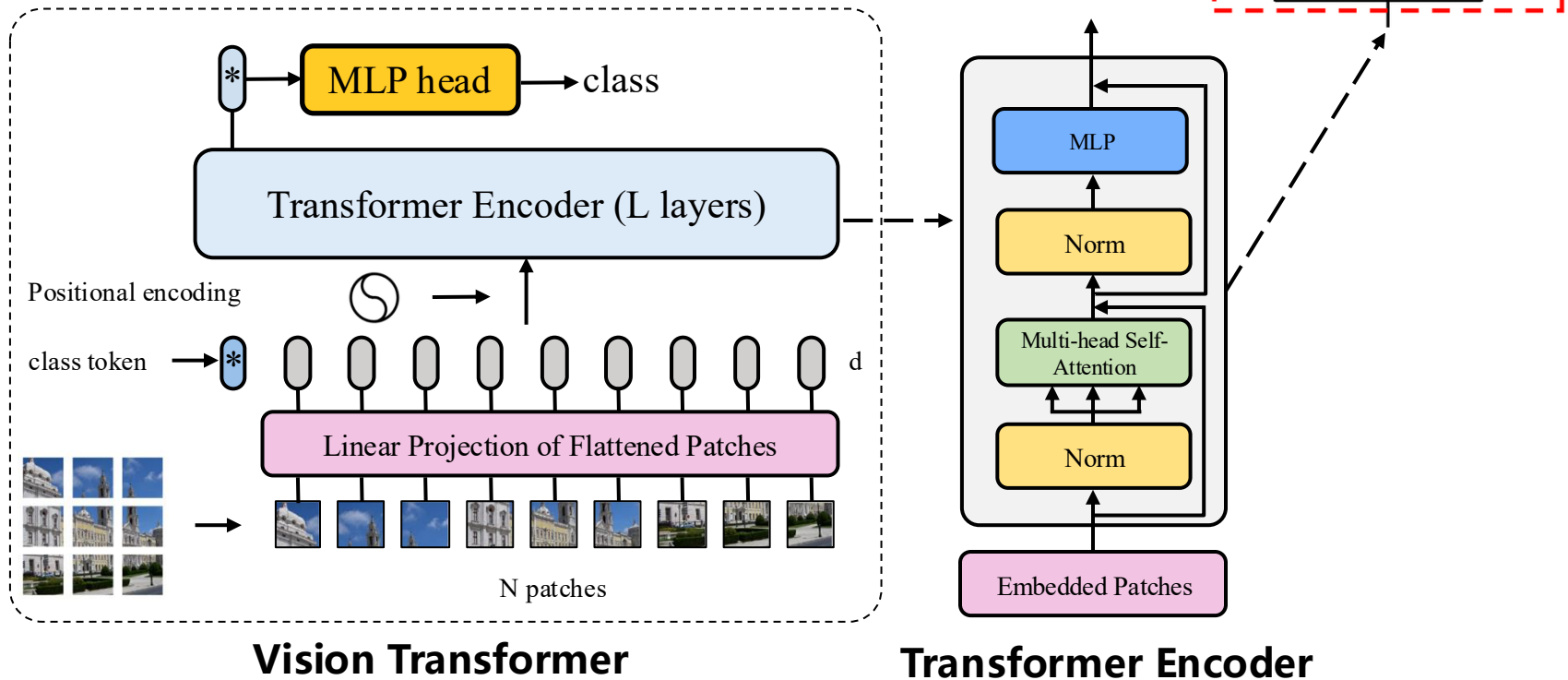


Quick Review

ViT, Mamba, Vision Mamba

Vision Transformer

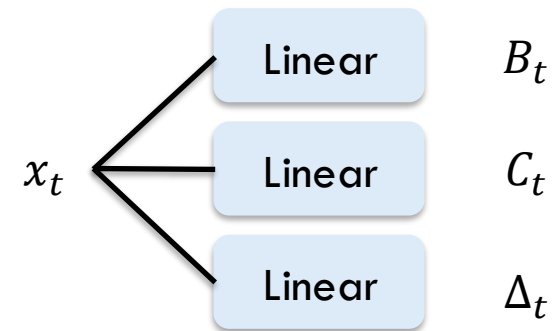
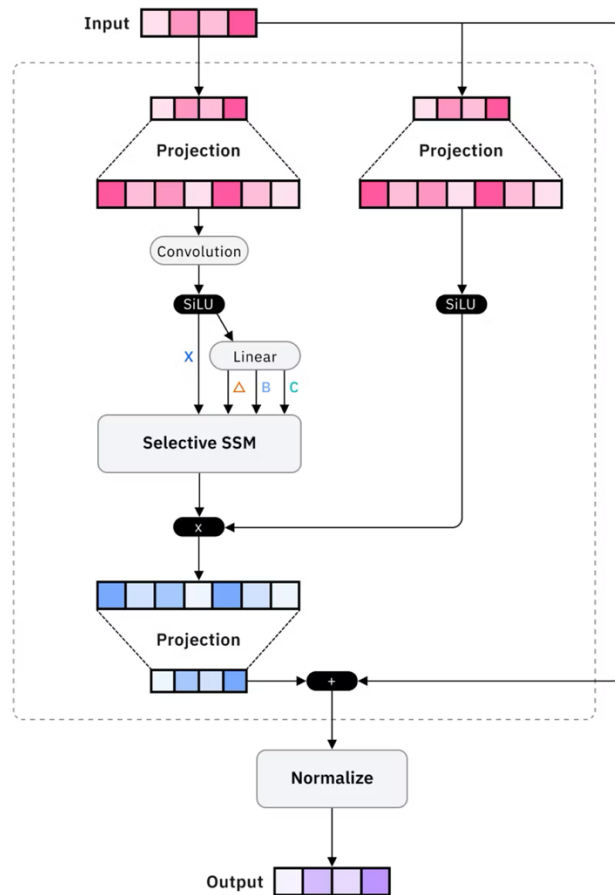
Overall Framework



Dosovitskiy et al., An Image is Worth 16*16 Words: Transformers for image recognition, ICLR, 2021, Google Brain

Mamba – Selective State Space Models (S6)

Mamba Block



Algorithm 2 SSM + Selection (S6)

Input: $x : (B, L, D)$

Output: $y : (B, L, D)$

1: $A : (D, N) \leftarrow \text{Parameter}$

▸ Represents structured $N \times N$ matrix

2: $B : (B, L, N) \leftarrow s_B(x)$

3: $C : (B, L, N) \leftarrow s_C(x)$

4: $\Delta : (B, L, D) \leftarrow \tau_\Delta(\text{Parameter} + s_\Delta(x))$

5: $\bar{A}, \bar{B} : (B, L, D, N) \leftarrow \text{discretize}(\Delta, A, B)$

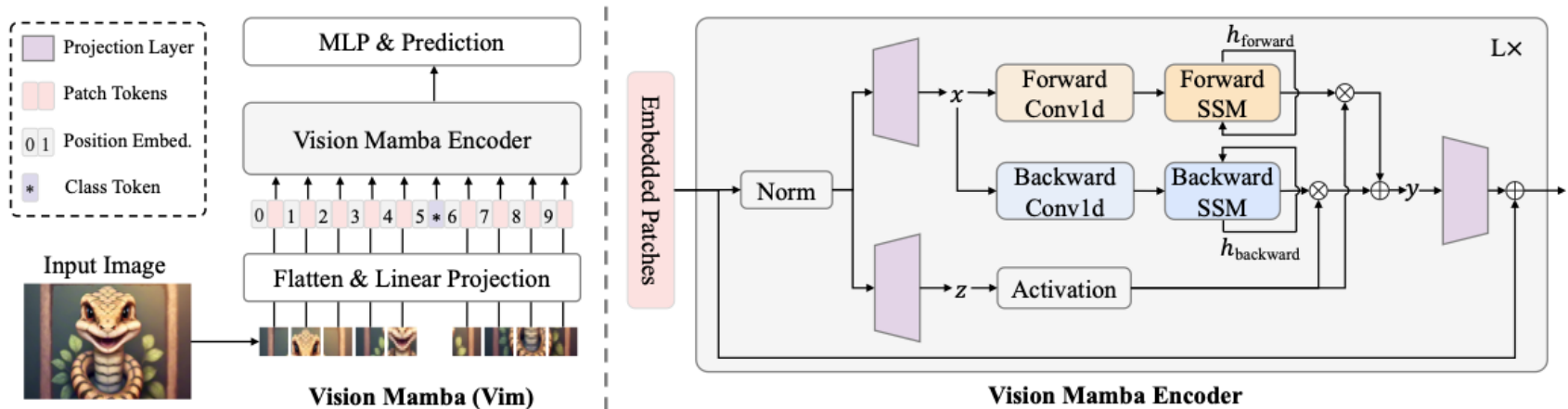
6: $y \leftarrow \text{SSM}(\bar{A}, \bar{B}, C)(x)$

▸ **Time-varying:** recurrence (*scan*) only

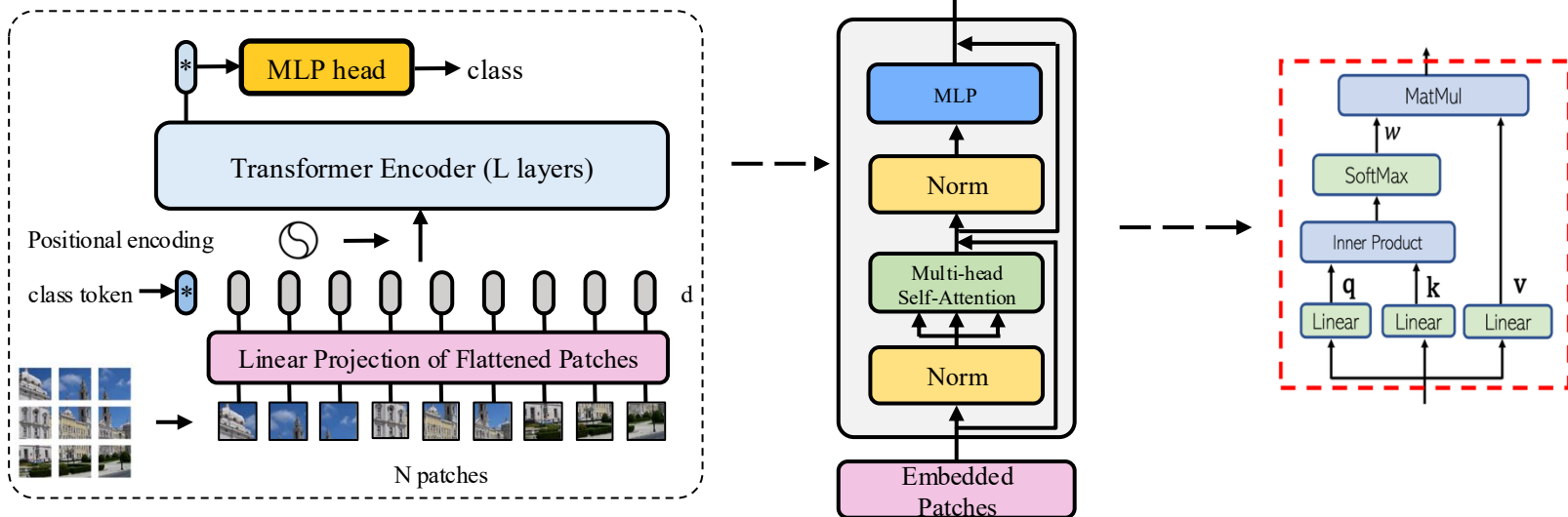
7: **return** y

Vision Mamba and Vision Transformer

Vision Mamba



Vision Transformer



Large-Language Model

LLM & MLLM: Emerging Capabilities

I was prescribed a statin tell me about statins and what questions should I ask at my follow up

Searching for information on statin interactions and guidelines
Get a quick answer >

Questions to bring to your follow-up

1. **Pregnancy/breastfeeding plans (if relevant):** What's the plan? Mayo Clinic

2. If I can't reach our LDL goal or can't tolerate can include ezetimibe or PCSK9-targeting t
Guideline-based add-ons are considered w
threshold on maximally tolerated statin. Mayo Clinic
Statin side effects: Weigh the benefits and risks – Mayo Clinic
July 20, 2025 — Statins block an enzyme the liver needs to make cholesterol. This cause...

3. Interactions/diet: Is grapefruit okay with my supplements I should avoid or separate in time (e.g., certain antibiotics/antifungals, HIV meds, or large doses of niacin)? U.S. Food and Drug Administration Cleveland Clinic

It looks like the plant in your photo is a cactus, not a *snake plant* — so if your snake plant is turning yellow, it's not the one pictured here.

Yellowing leaves on a snake plant can happen for a few common rea

Why does my snake plant look yellow?

1. **Overwatering (Mo**
Snake plants store wa +
don't like sitting in soggy soil. Too much moisture can lead to root rot, which first shows as yellow, mushy leaves.

- Fix: Let the soil dry out completely before watering again. Use a well-draining potting mix and a pot with drainage holes.

2. While hardy, snake plants still need occasional water. Long droughts can cause

Dreamscape Tracker

```
1 import React, { useState } from 'react';
2 import { Button } from "@components/ui/button";
3 import { Card, CardContent } from "@components/ui/ca
4 import { Input } from "@components/ui/input";
5 import { Textarea } from "@components/ui/textarea";
6 import { motion } from "framer-motion";
```

Create an app that allows me to track my dreams, while generating a soundscape and image for each dream

Colors Beyond the Horizon

0:40 7:25

16 };
17 };
18 const generateImage = async (dreamText) => {
19 // Placeholder: Integrate with image generation API
20 return 'https://example.com/image/\${encodeURIComponent(dreamText)}';
21 };
22 }

write me a wedding toast for my friends giselle and tyler based on this attachment of notes

Ask ChatGPT

Thought for 7s

Good evening everyone—I'm lucky to be one of Giselle's friends, and even luckier to be here for this.

Good evening everyone—I'm lucky to be one...

Can you make the introduction more engaging?

write me a wedding toast for my friends giselle and tyler based on this attachment of notes

Thought for 7s

Good evening everyone—I'm lucky to be one of Giselle's friends, and even luckier to be here for this.

Can you make the introduction more engaging?

write me a wedding toast for my friends giselle and tyler based on this attachment of notes

Thought for 7s

Good evening everyone—I'm lucky to be one of Giselle's friends, and even luckier to be here for this.

Can you make the introduction more engaging?

LLM & MLLM: Emerging Capabilities

1

Input

A cartoon mail delivery person with a smile on her face. She is standing facing forward in front of a white background.

2

Output



2

Input

Here, Sally is about to deliver a letter. Sally is standing in front of a red door to a house, holding a letter in her hand. We are looking at her from the side.

3

Output



4

Input

Now Sally is being chased by a dog. Sally is running down the sidewalk and as a golden retriever is chasing her.

5

Output



6

Input

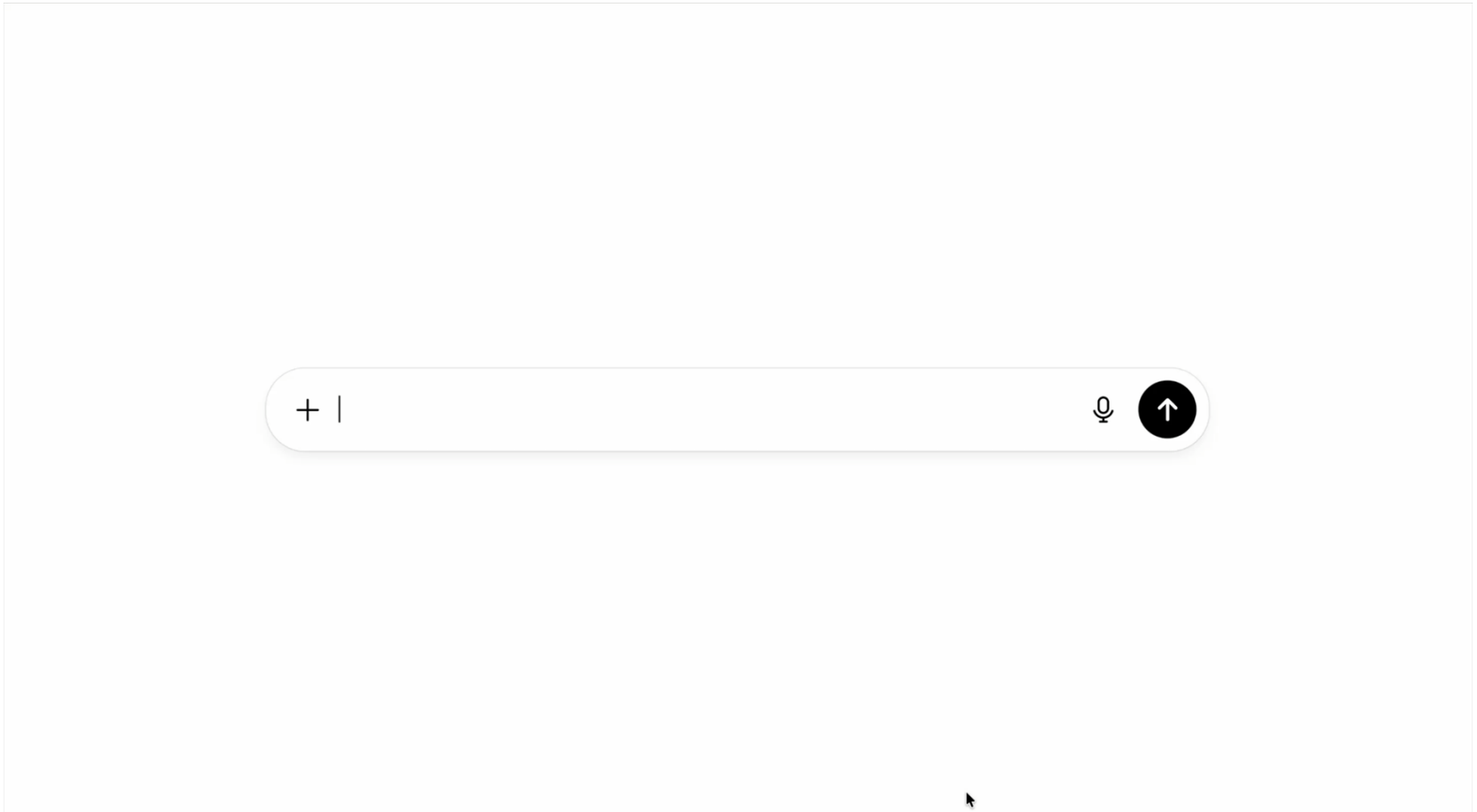
Uh oh, Sally has tripped! Sally has tripped over a branch that was blocking the sidewalk, and she is trying to stand up. The dog is still chasing her in the background.

7

Output



LLM & MLLM: Emerging Capabilities



+ |  

What is Large Language Model (LLM)?



Large Language Model (LLM) is the most representative model of GenAI designed to process and generate human-like text. It uses vast amounts of data and sophisticated algorithms to understand, predict, and create natural language. -- *Generative by ChatGPT-4o*

Text Prompt:
What is Large Language model?

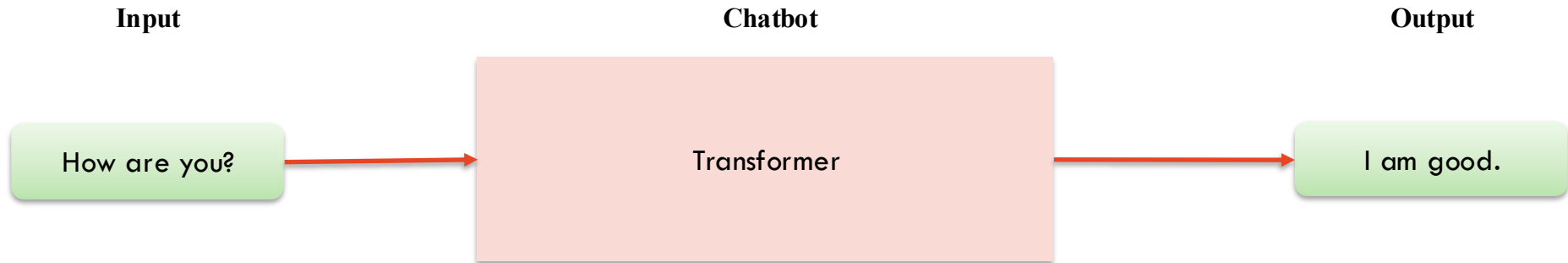


Large Language Model

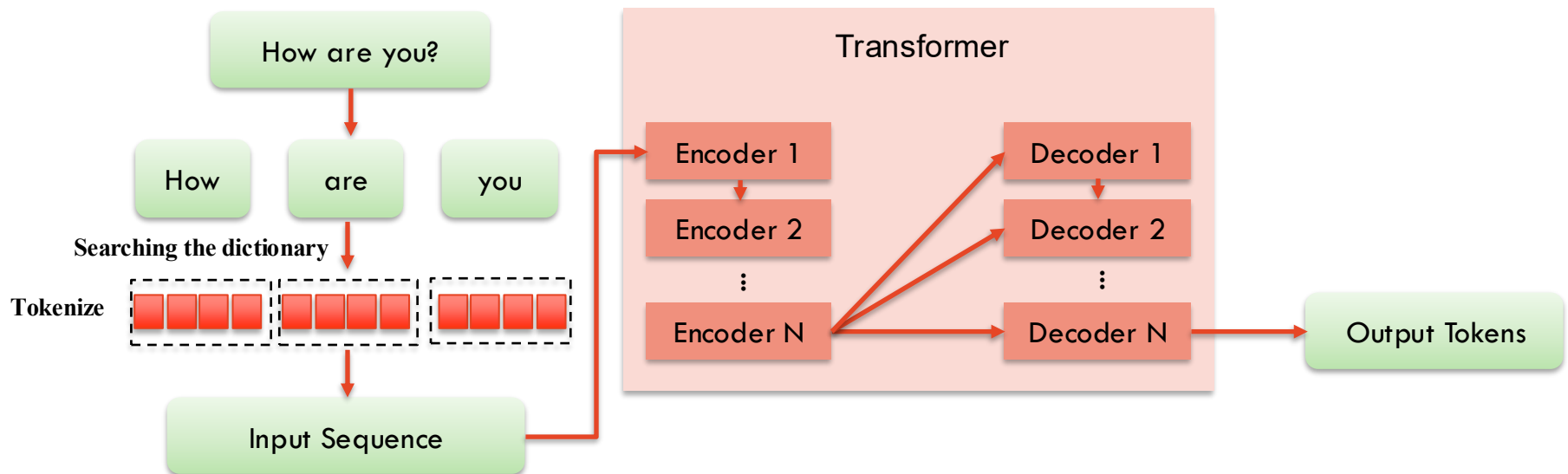


LLM is the most

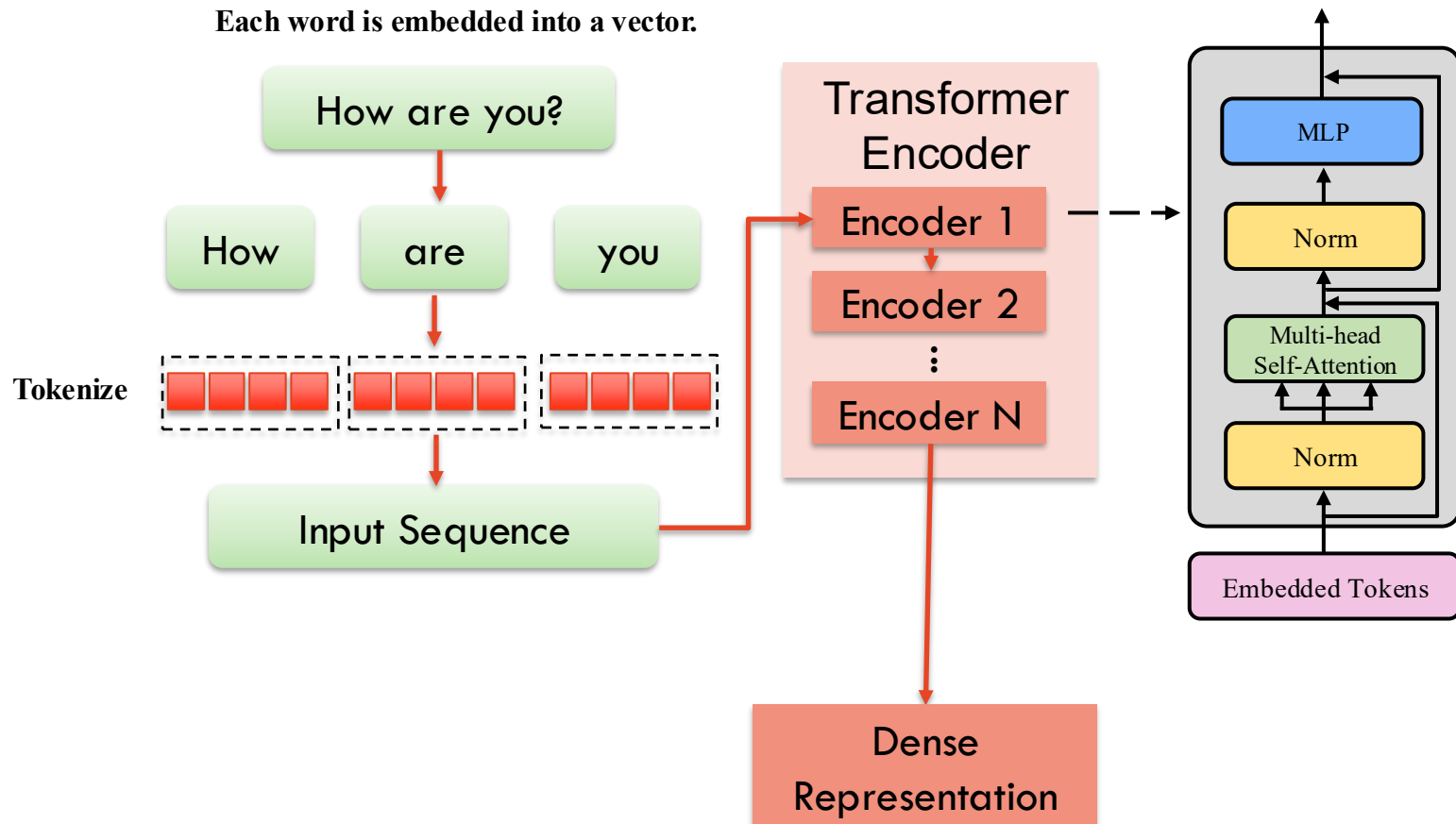
Transformer Architecture



Each word is embedded into a vector.

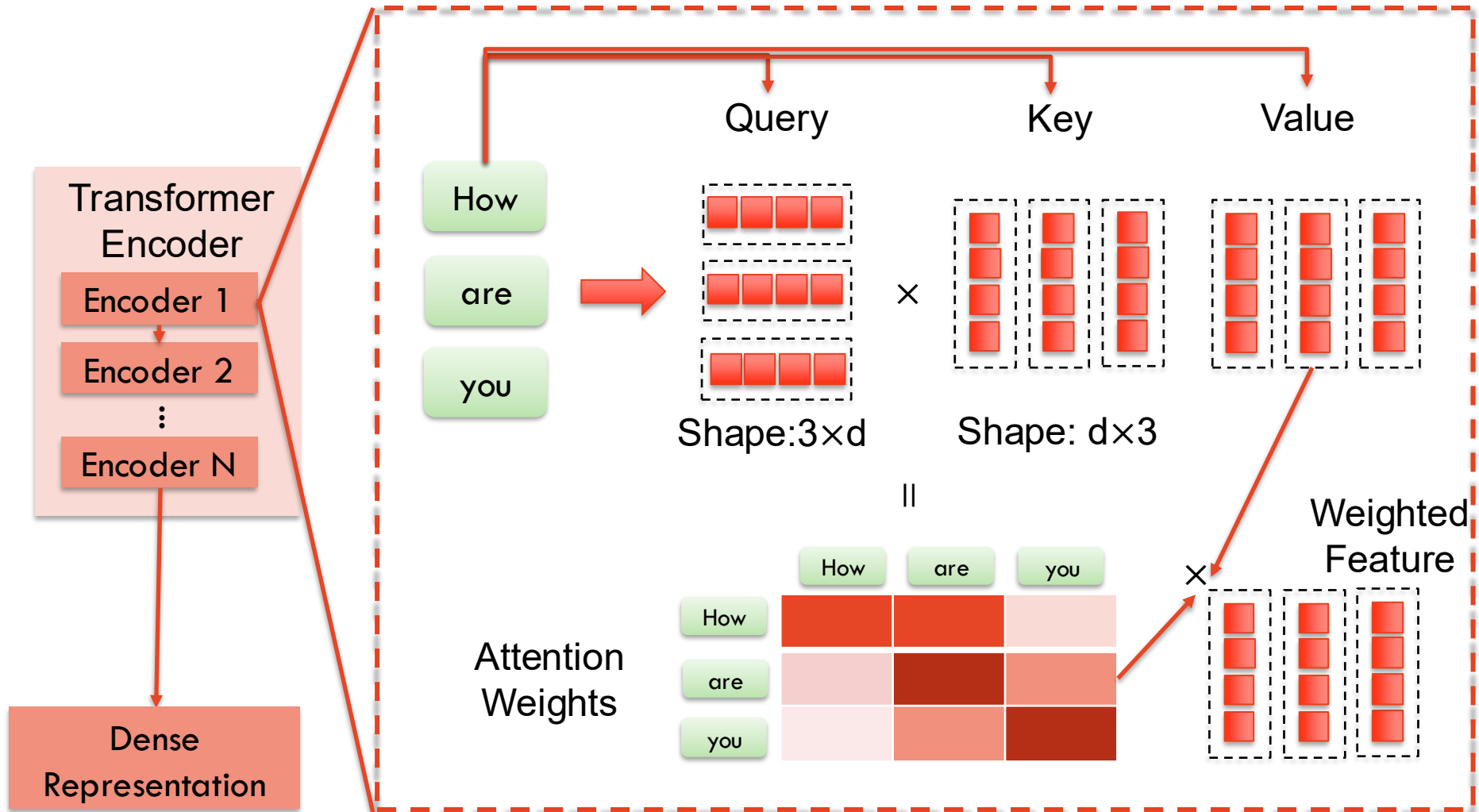


Transformer Encoder

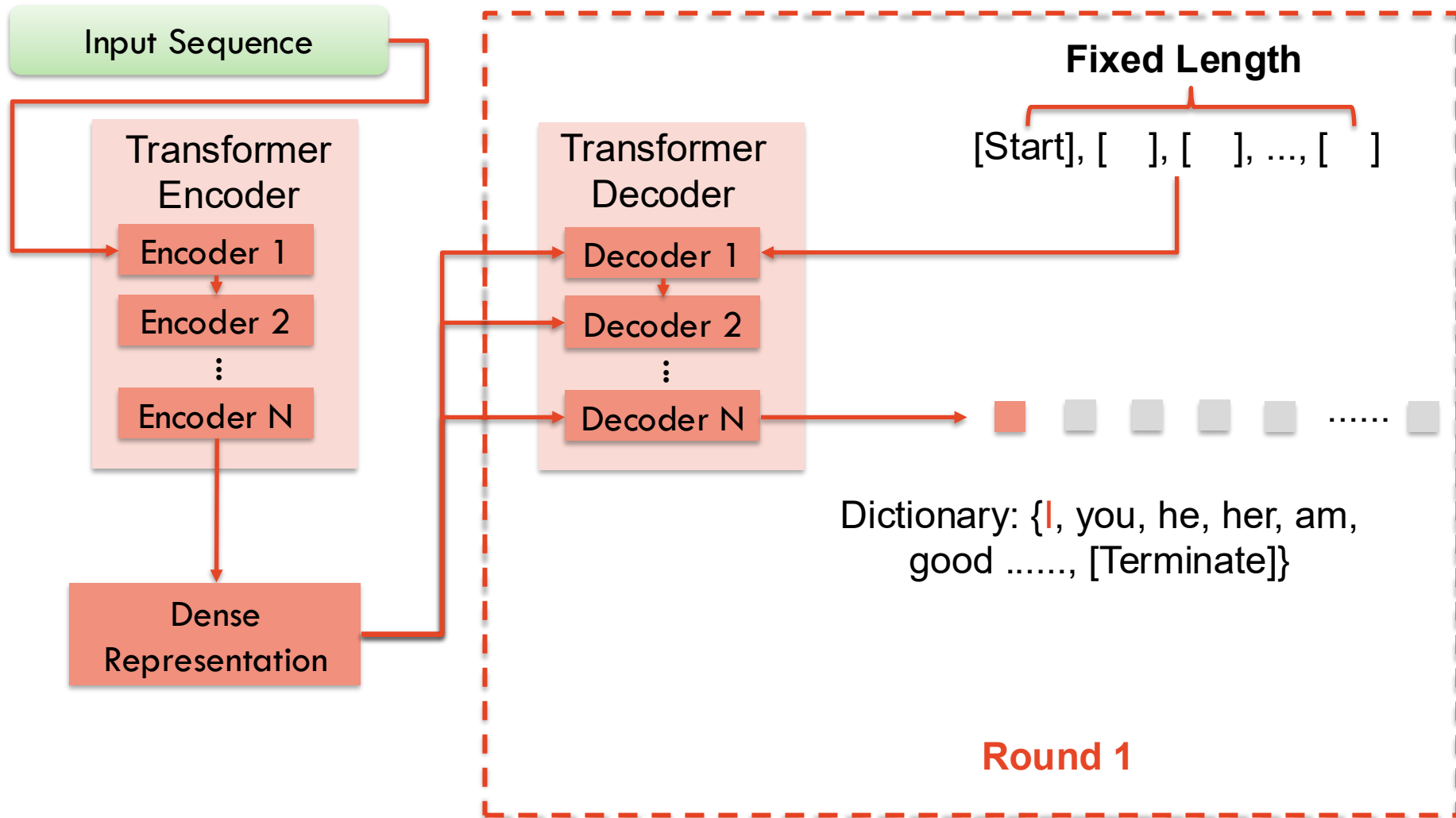


It can be used to downstream tasks like sentiment analysis.

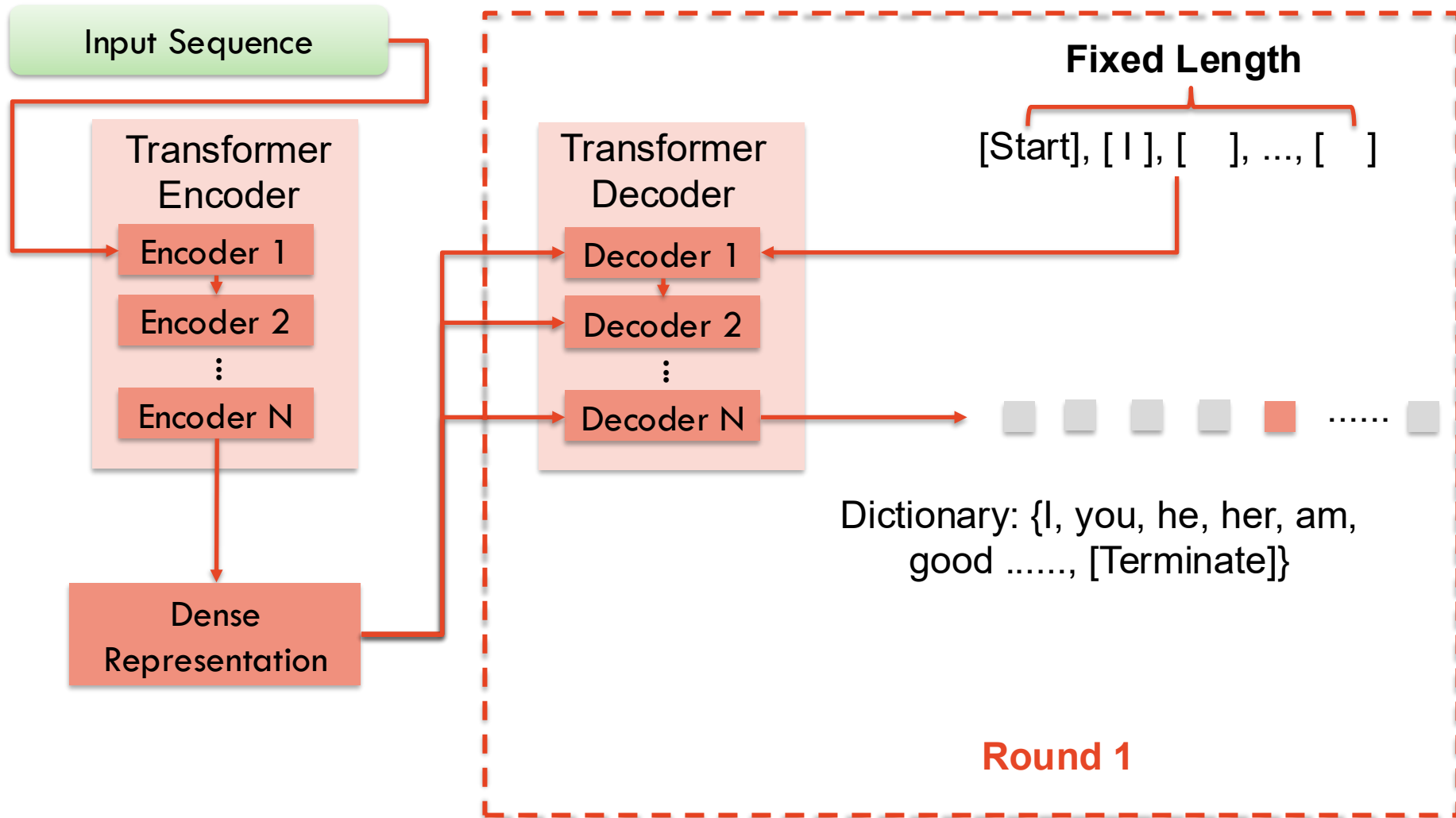
Attention



Transformer Decoder – Autoregressive Generation



Transformer Decoder – Autoregressive Generation

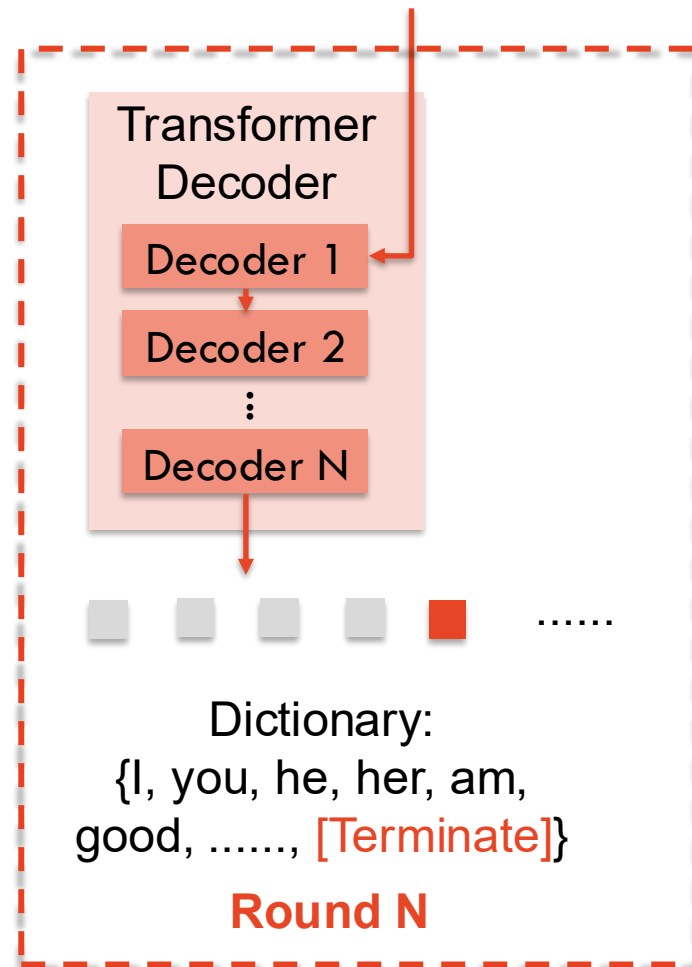
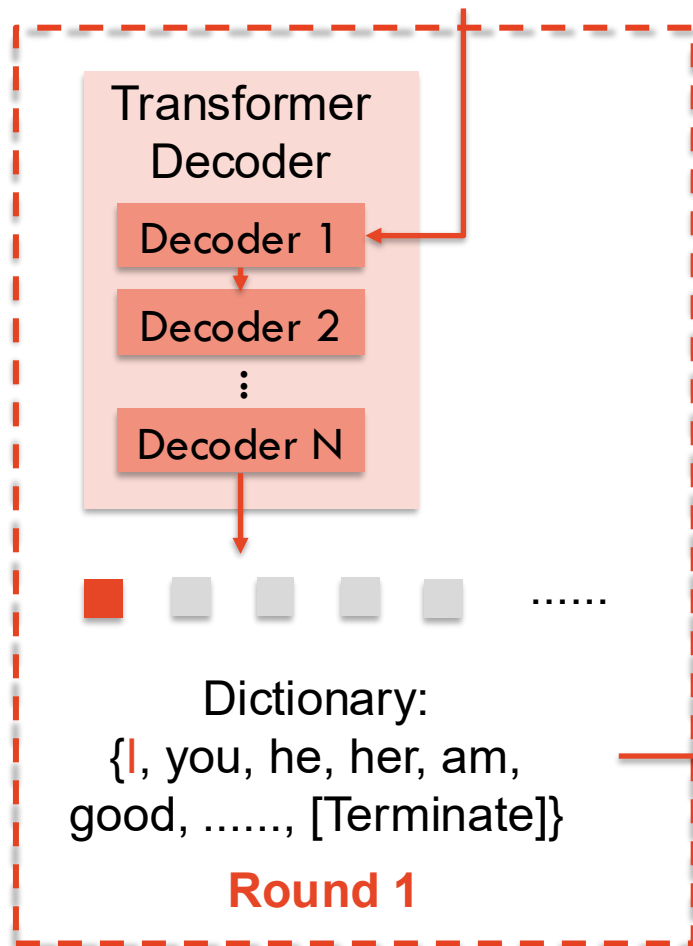


Transformer Decoder Only – How GPT works

Prompt:

Please, response, to, How, are, you, [Start]

Please, response, to, How, are,
you, [Start], I, am, good, [Terminate]



Decoder Only v.s. Encoder-Decoder

Autoregressive Models

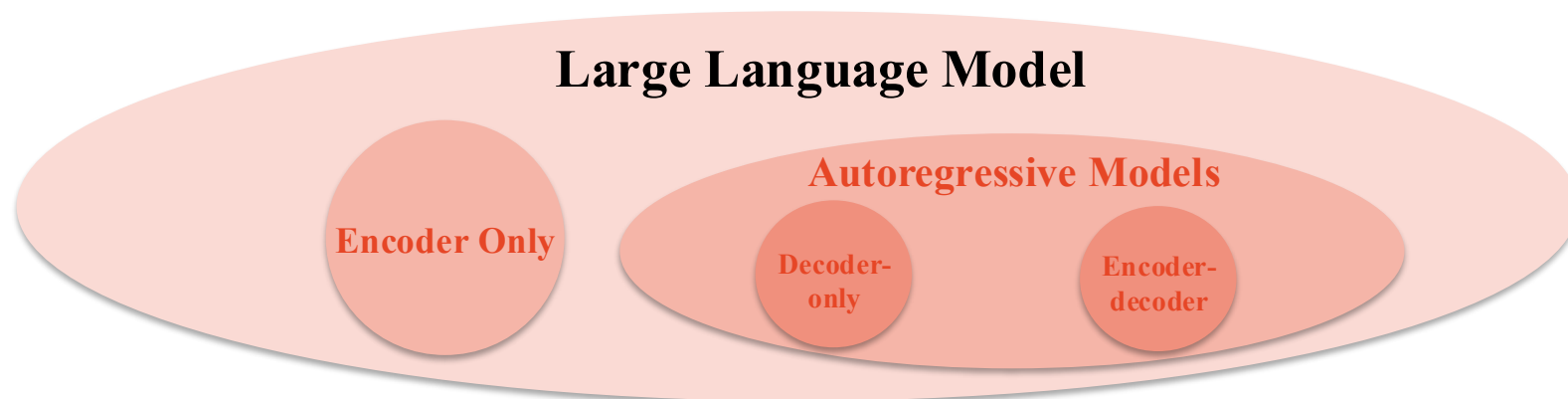
- *Like predictive texting, where each next word is based on what you've written so far.*

Decoder-only Architecture

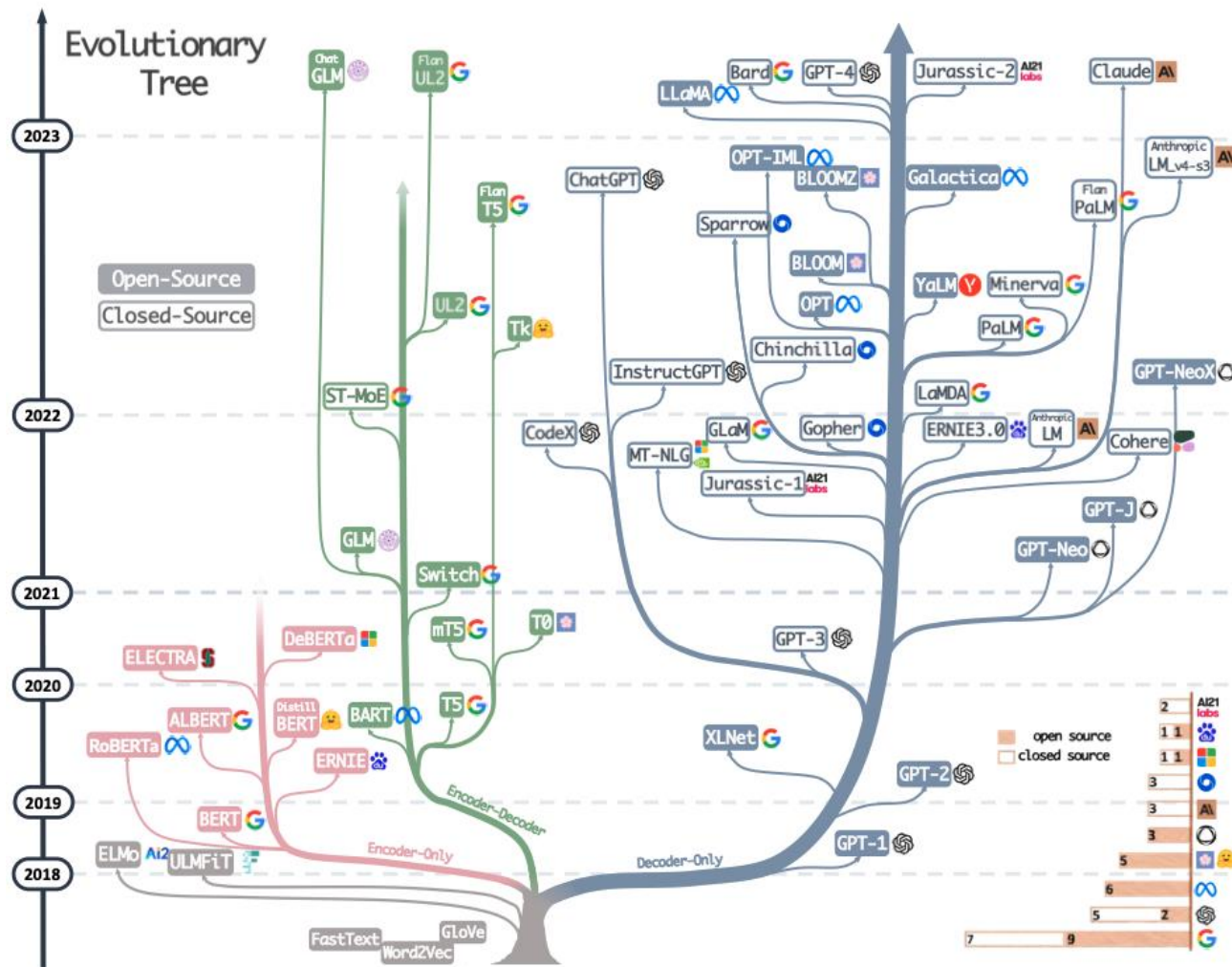
- *Like writing a story one word at a time, only considering what you've already written*

Encoder-Decoder Architecture

- *Like translating a sentence from one language to another—understand the full meaning first (encoder) and then write the translation (decoder).*

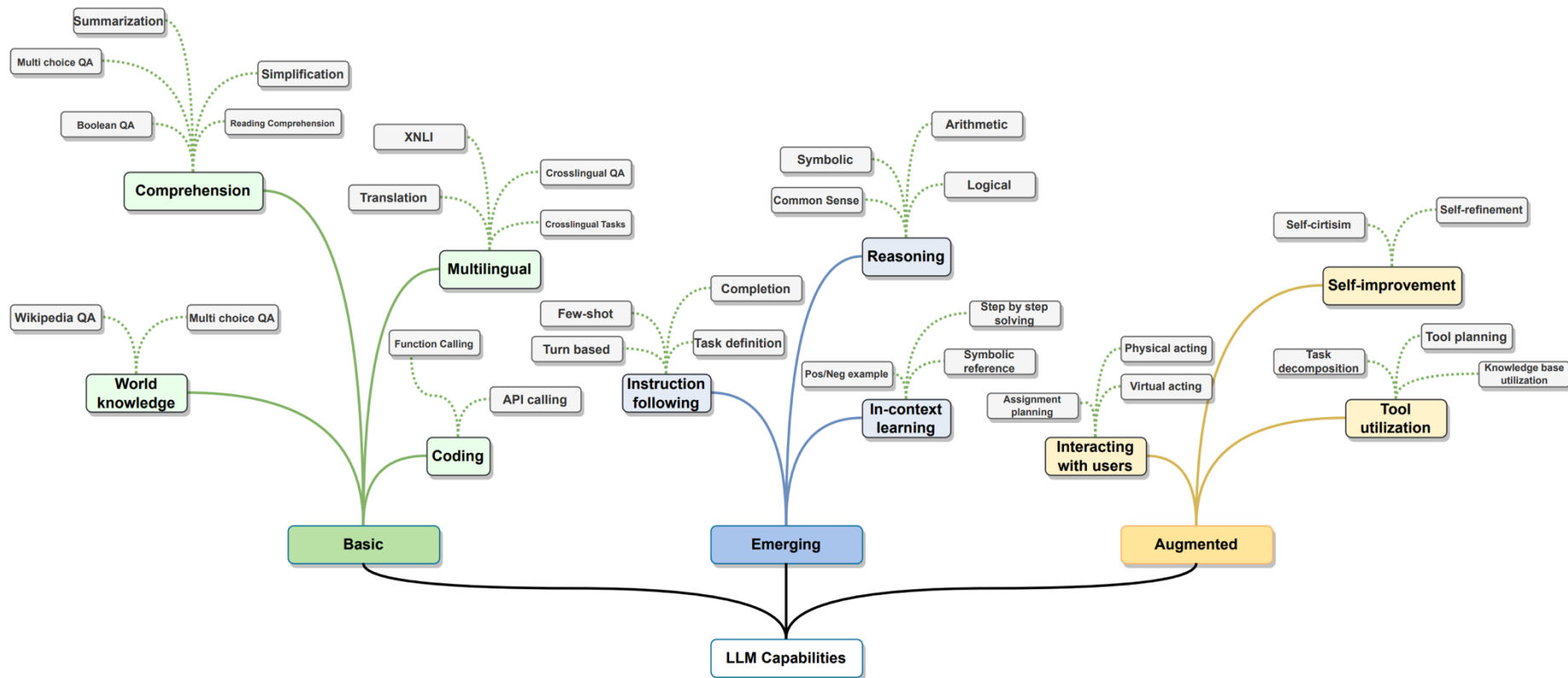


Development of LLMs



Yang et al., Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond

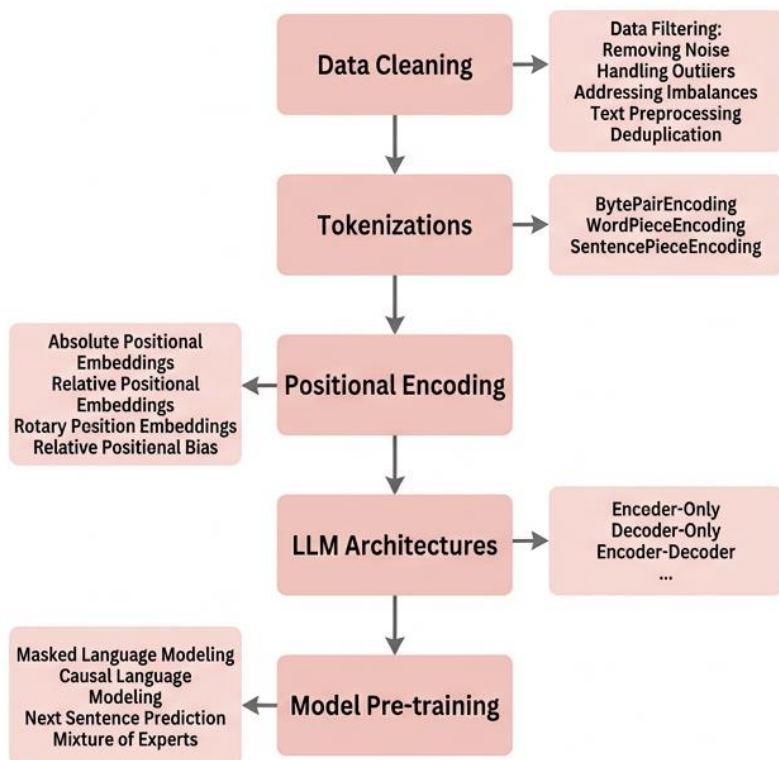
LLM Capabilities



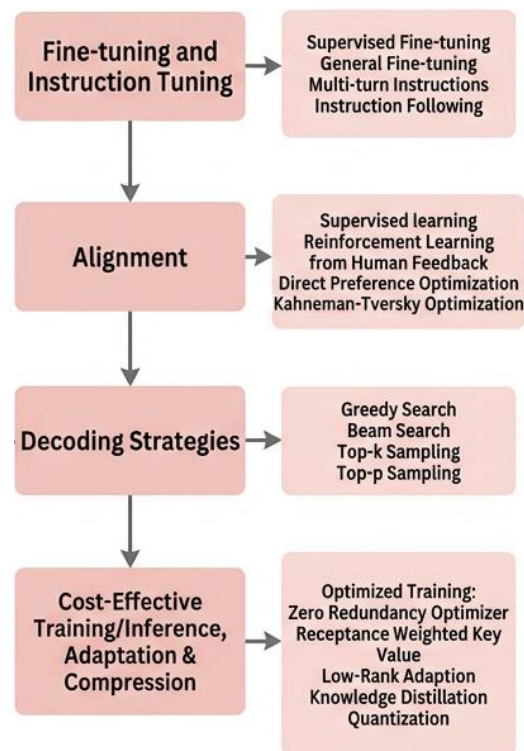
Minaee S, Mikolov T, Nikzad N, et al. Large language models: A survey

LLMs Training Pipeline Overview

How LLMs Are Built? (Part 1)



How LLMs Are Built? (Part 2)



Minaee S, Mikolov T, Nikzad N, et al. Large language models: A survey

Multi-modal Language Model

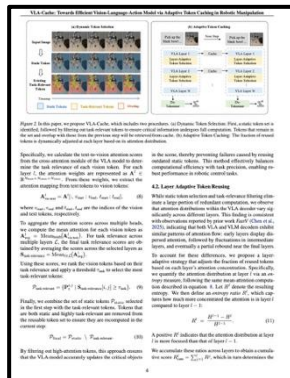
From LLM to VLM



Large Language Model




This document
talks about ...



Large Language Model




Sorry, I can not
understand..



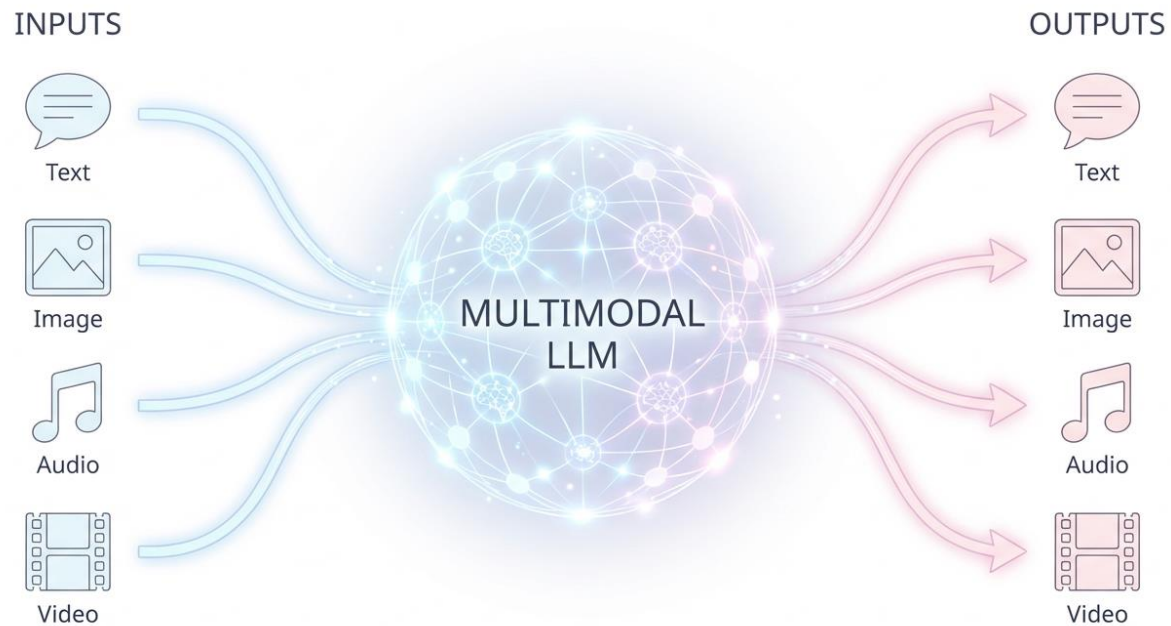
Multi-Modal Large Language Models (MLLMs)

MLLMs enable LLMs to comprehend and generate content across diverse modalities

- *Image*
- *Video*
- *Audio*
- ...

LLMs

- *Text Input Only*



Generated by Gemini

MLLMs Architecture

Vision Encoder

- *ViT / CLIP / Conv-based encoders*
- *Extract visual and auditory representations from images or video frames*

Multimodal Alignment Module

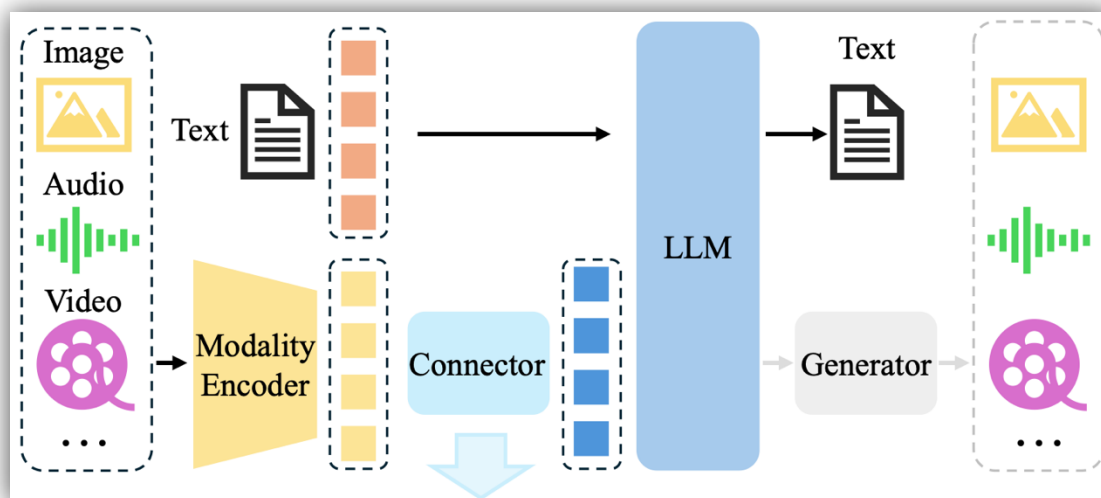
- *Projection layers / Q-Former / Cross-attention*
- *Align visual features with language embeddings*

LLM Backbone

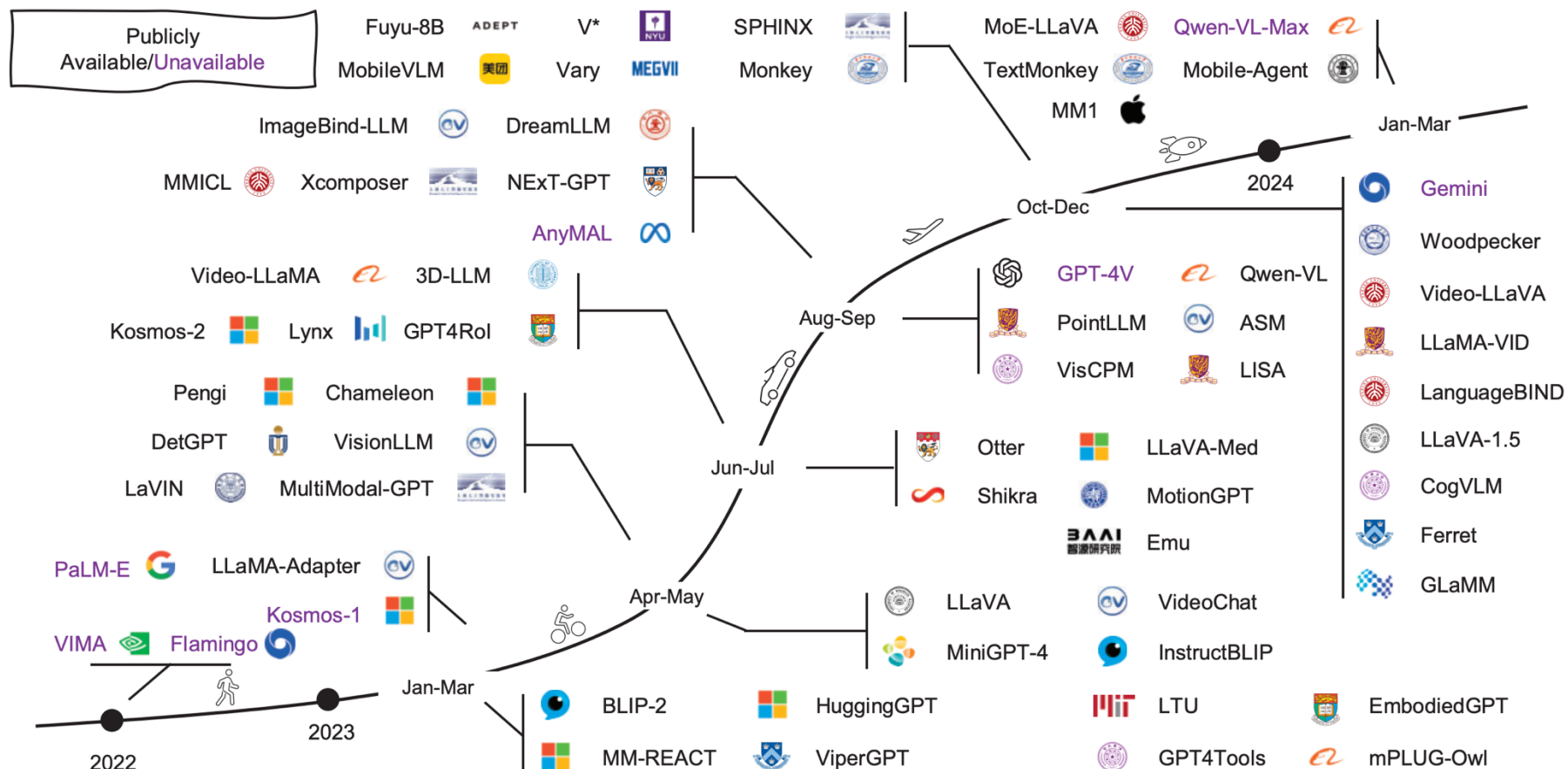
- *Transformer-based LLM*
- *Performs reasoning and generates language outputs*

Generator (Optional)

- *Produces modality-specific outputs beyond text (for LLMs limited to text generation)*



Development of MLLMs



Yin S, Fu C, Zhao S, et al. A survey on multimodal large language models

Pretrained Encoder and LLMs

Image Encoders

Variants	Pre-training corpus	Resolution	Samples (B)	Parameter size (M)
OpenCLIP-ConvNext-L [33]	LAION-2B	320	29	197.4
CLIP-ViT-L/14 [12]	OpenAI's WIT	224/336	13	304.0
EVA-CLIP-ViT-G/14 [34]	LAION-2B, COYO-700M	224	11	1000.0
OpenCLIP-ViT-G/14 [33]	LAION-2B	224	34	1012.7
OpenCLIP-ViT-bigG/14 [33]	LAION-2B	224	34	1844.9
InternViT-6B [35]	Multiple datasets	448	–	5540.0

Open-sourced LLMs

Model	Release date	Pre-train data scale	Parameter size (B)	Language support	Architecture
Flan-T5-XL/XXL [44]	Oct. 2022	–	3/11	En, Fr, De	Encoder decoder
LLaMA [45]	Feb. 2023	1.4T tokens	7/13/33/65	En	Causal decoder
Vicuna [46]	March 2023	1.4T tokens	7/13/33	En	Causal decoder
LLaMA-2 [47]	July 2023	2T tokens	7/13/70	En	Causal decoder
Qwen [48]	Sept. 2023	3T tokens	1.8/7/14/72	En, Zh	Causal decoder
LLaMA-3 [49]	April 2024	15T tokens	8/70/405	En, Fr, De, etc.	Causal decoder

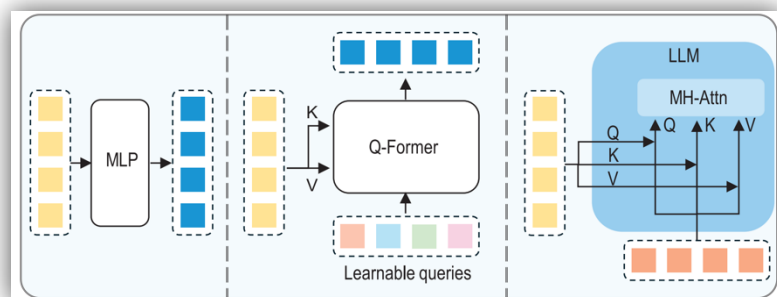
Connector - Modality Interface

Connector aligns multimodal features with the LLMs

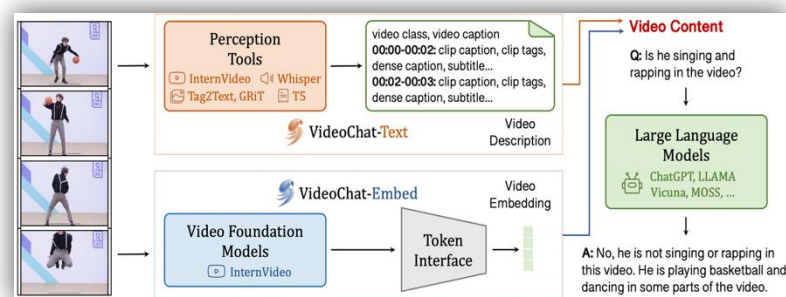
Learnable connector

- *Token-level fusion*
- *Feature-level fusion*

Expert-to-text translation



Learnable connector (trainable alignment module)



Expert-to-text translation (no training, but lossy)

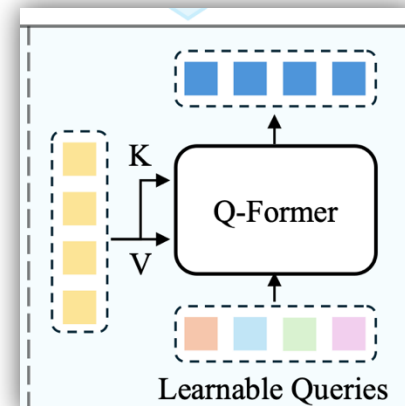
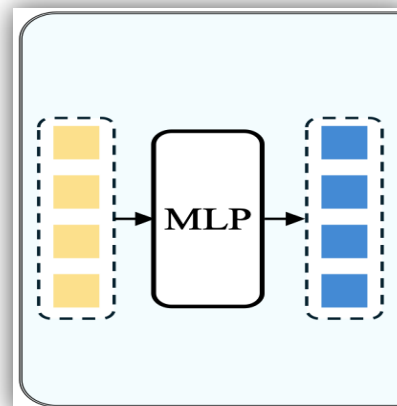
Li K C, He Y, Wang Y, et al. Videochat: Chat-centric video understanding

Learnable Connector

Token-level fusion

convert features $\rightarrow$ visual tokens $\rightarrow$ concat with text

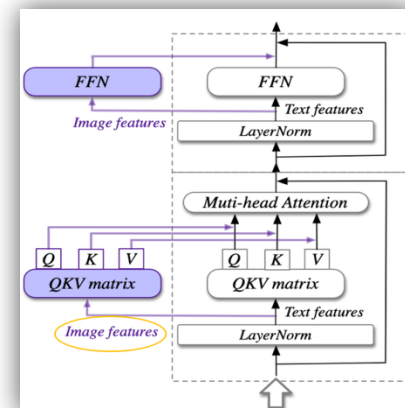
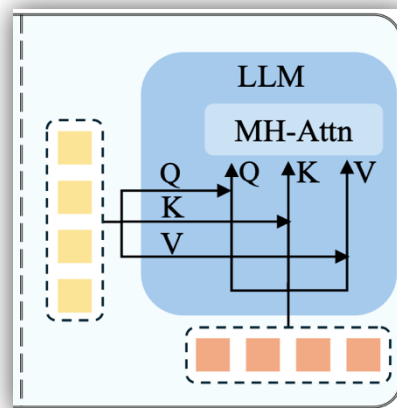
- *Projection/MLP (e.g., LLaVA)*
- *Query/Q-Former (e.g., BLIP-2)*



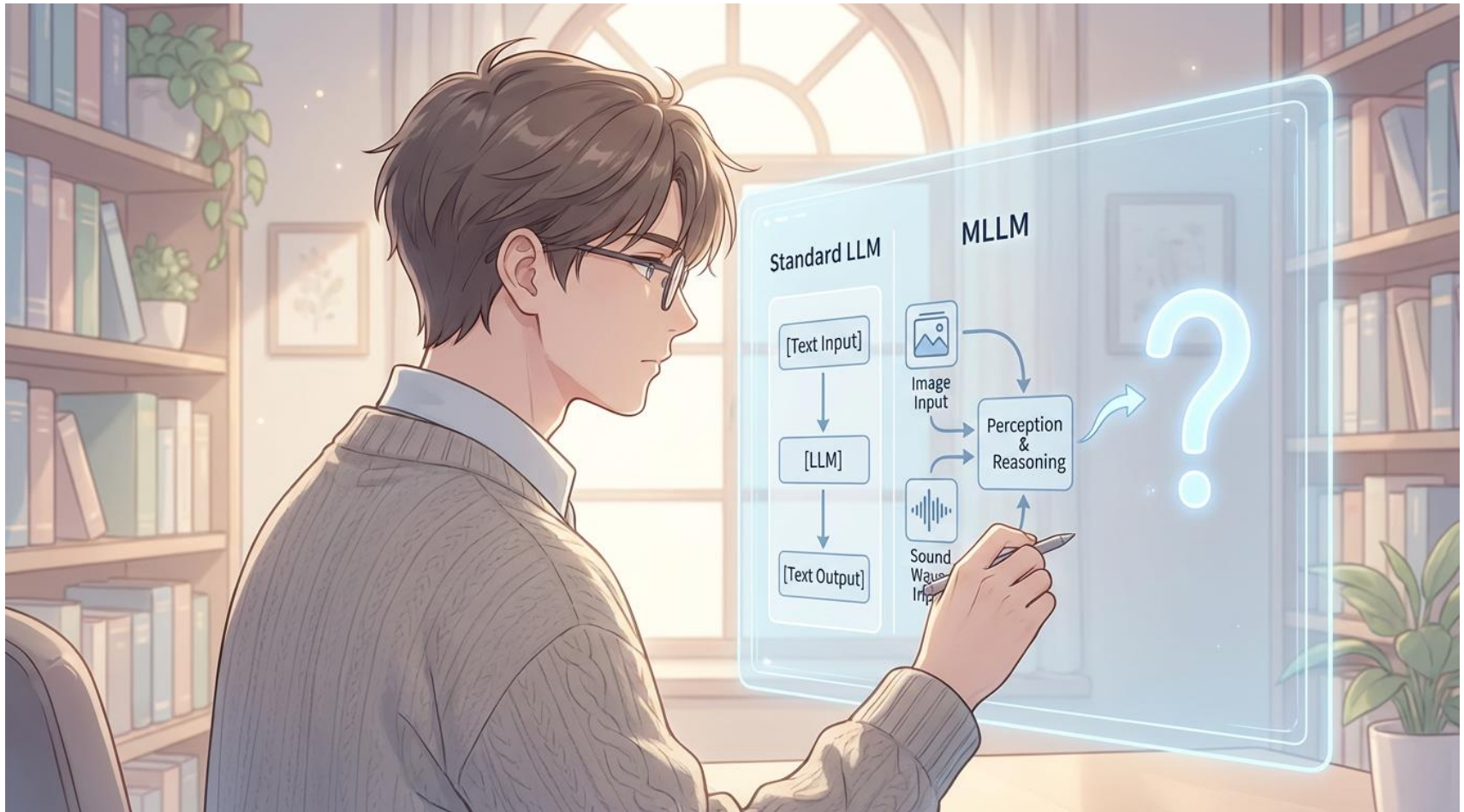
Feature-level fusion

inject modules for deep cross-modal interaction

- *Cross-attn insertion (e.g., Flamingo)*
- *Layer-wise visual expert / adapters (e.g., CogVLM, LLaMA-Adapter)*



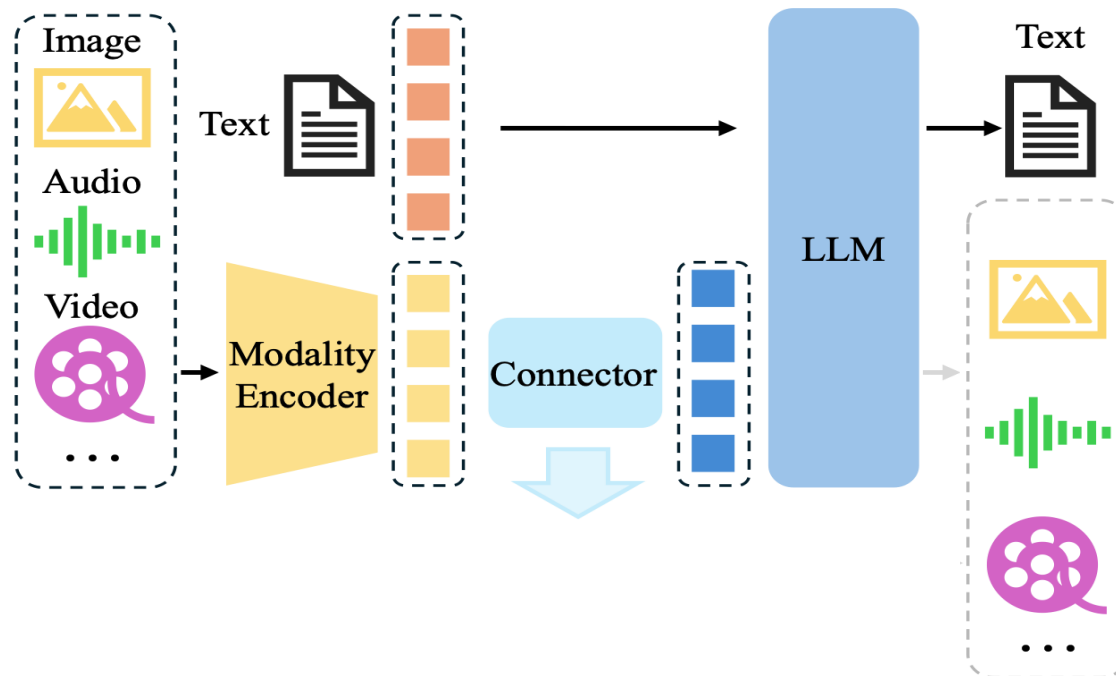
How can MLLMs Generate Non-text Modalities?



Using MLLMs to Generate Multi-Modal Data

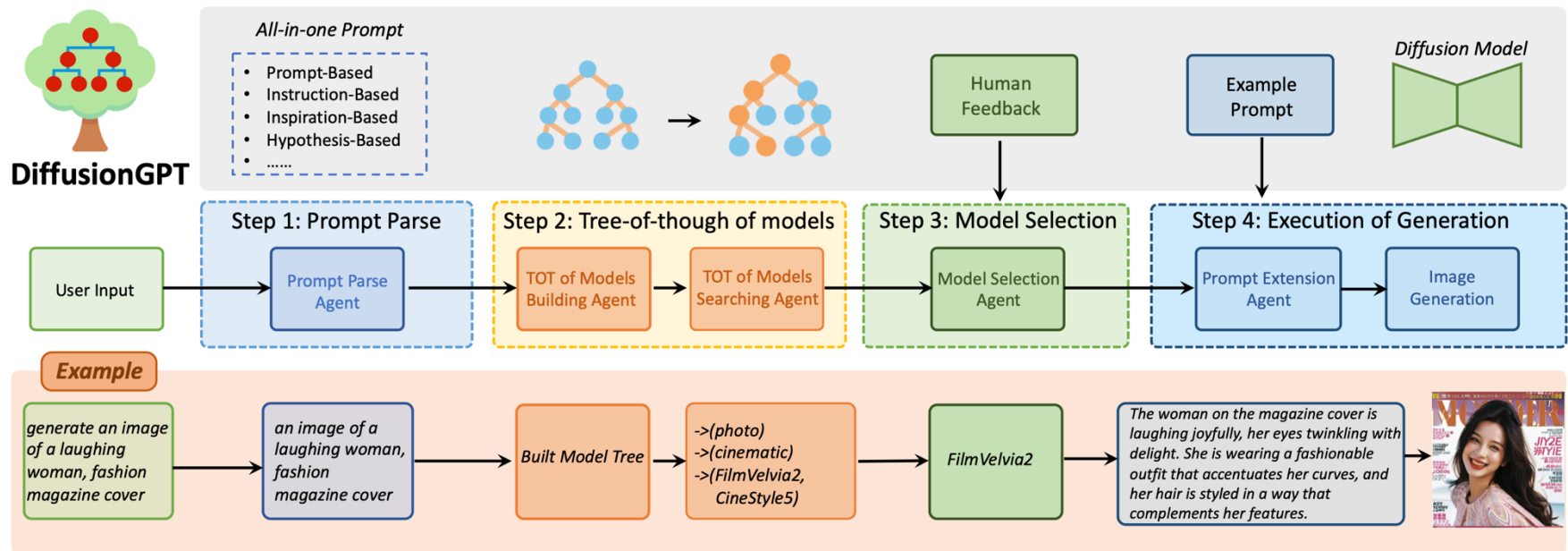
LLMs as Conditioner

- *Diffusion GPT, ThinkDiff*



LLMs as Conditioner - DiffusionGPT

- Use LLMs as a unified controller for diffusion-based image generation
- Combine prompt understanding with **expert model selection**



Qin J, Wu J, Chen W, et al. Diffusiongpt: Llm-driven text-to-image generation system

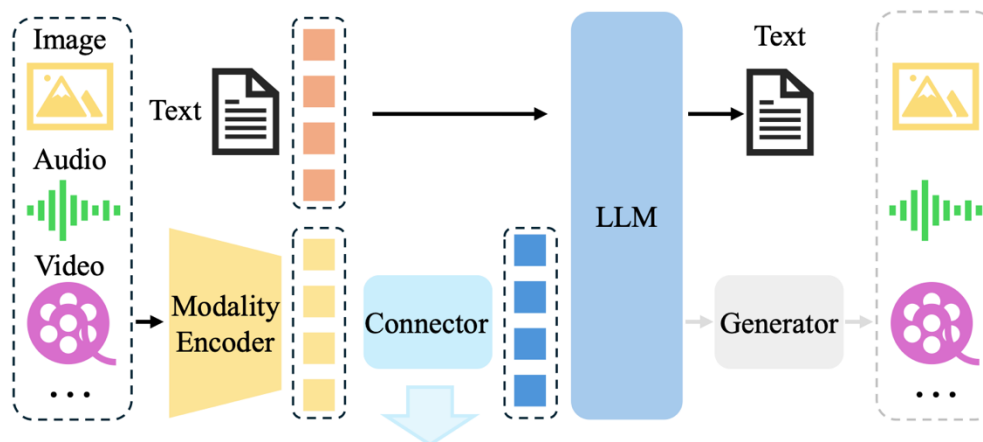
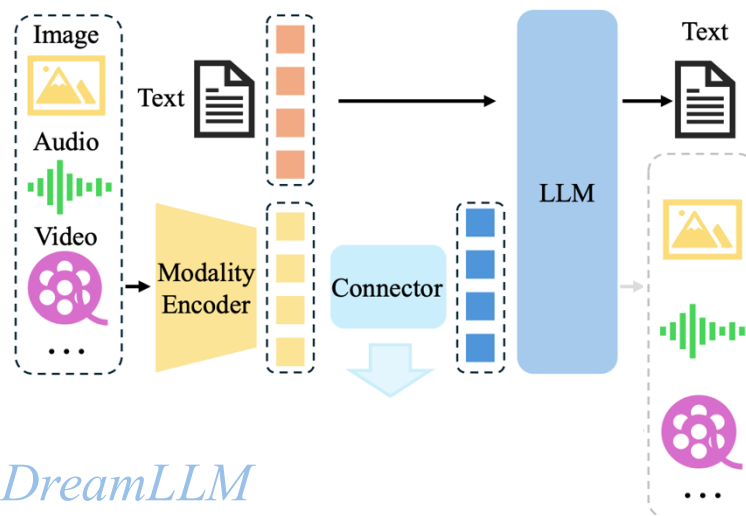
Using MLLMs to Generate Multi-Modal Data

LLMs as Conditioner

- *Diffusion GPT, ThinkDiff*

LLMs as Generator

- Visual Autoregressive
 - *Unified IO, Chamelon, SEED, DreamLLM*
 - *Unified IO2, SEED-X, Lumina-mGPT*
- Visual Scale Autoregressive
 - *VAR, STAR*
- Visual Diffusion
 - *Transfusion, Show-o*



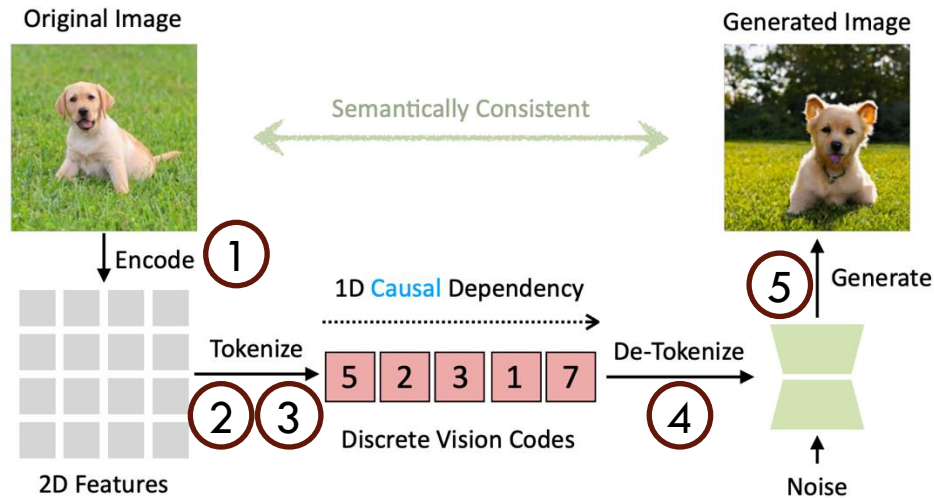
LLMs as Generator - Visual Autoregressive

MLLMs generate images using a visual autoregressive approach, similar to how LLMs generate text through next-token prediction.

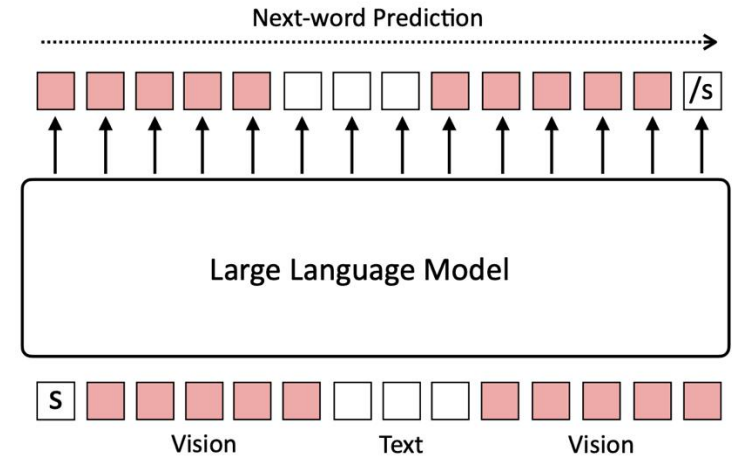
Key steps include:

- **Visual Vocabulary:** *Image features are discretized into a visual vocabulary (codebook) using VQ-GAN, transforming continuous features into discrete representations.*
- **Unified Learning:** *The unified learning process for text and images allows consistent training across modalities.*
- **Next-Token Prediction:** *Cross-entropy (CE) loss is used to predict the probability of the next token in the vocabulary.*

Visual Autoregressive - SEED (Tencent)



(a) SEED Visual Tokenizer



(b) Multimodal Autoregression with SEED tokens

- ① **ViT Encoder** (pre-trained): BLIP-2
- ② **Causal Q-Former**: Converts 2D image features into 1D features with causal dependency by ViT
- ③ **VQ Codebook**: Discretizes the 1D features before LLM
- ④ **Reverse Q-Former**: Maps the output of LLM to the condition of Stable Diffusion
- ⑤ **Unet Decoder** (pre-trained): Stable Diffusion

Visual Autoregressive - SEED-X (Tencent)

Tokenize: image $\rightarrow$ causal semantic tokens

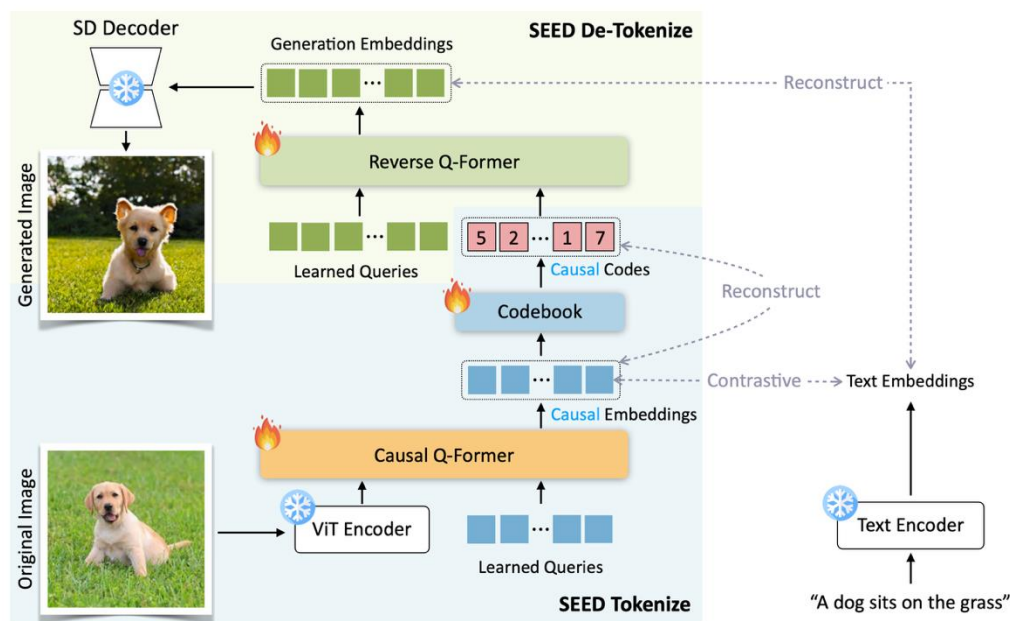
De-tokenize: tokens $\rightarrow$ generation embeddings

Training ensures tokens are:

- Causal
- Semantic
- Generative

Training Strategy

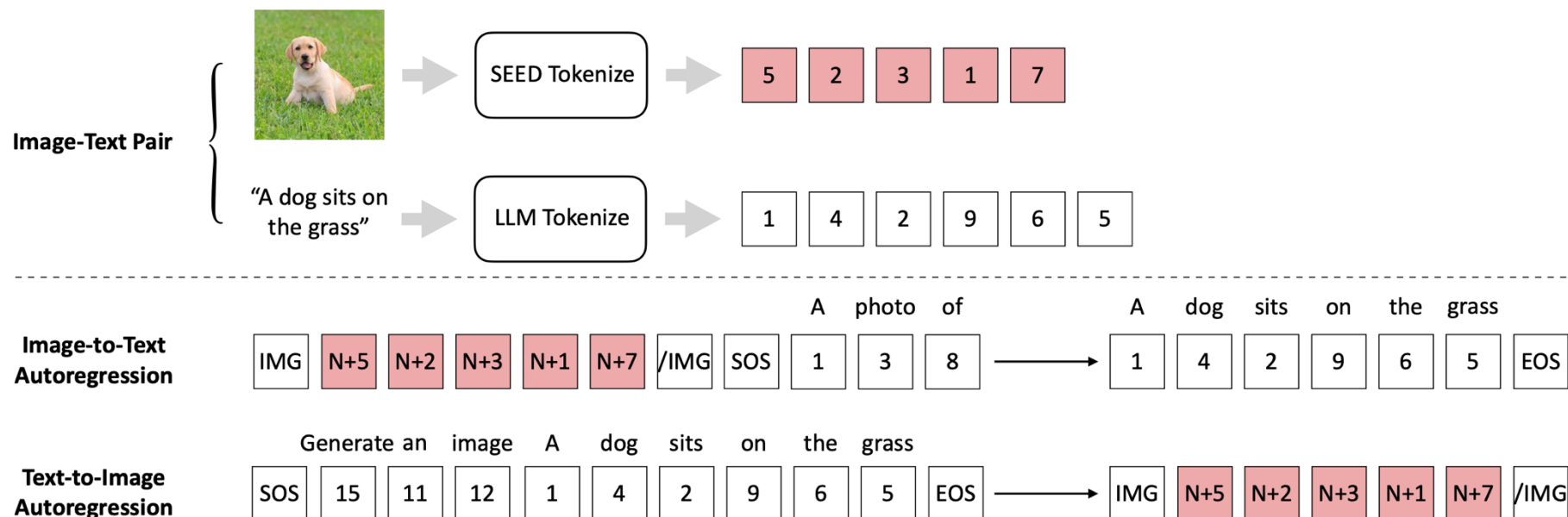
- **Causal Q-Former:** image–text contrastive learning
- **VQ + Reverse Q-Former:** dual reconstruction objectives



Visual Autoregressive - SEED-X (Tencent)

Two Autoregressive Tasks

- Image → Text: predict next text token
- Text → Image: predict next visual token



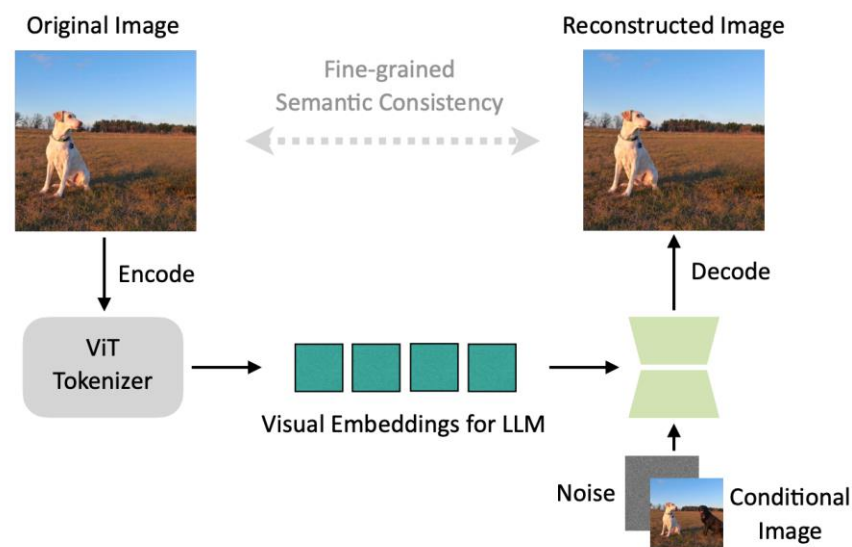
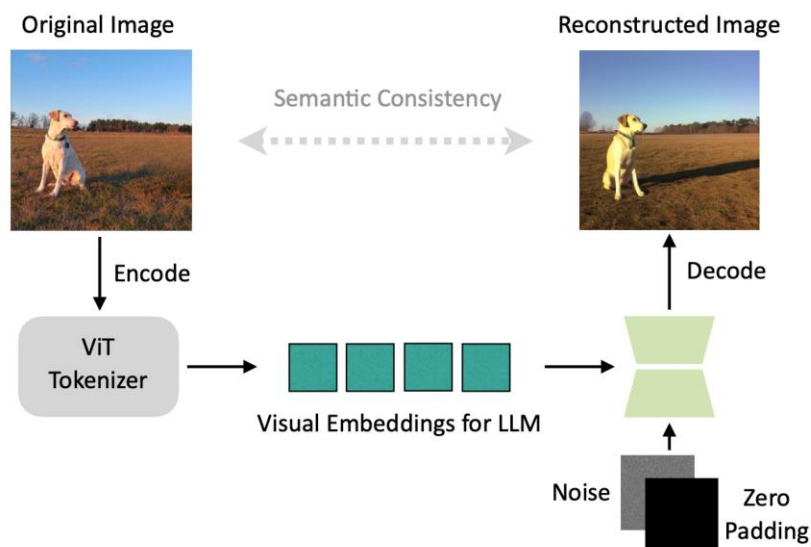
Visual Autoregressive - SEED-X (Tencent)

Use ViT features directly as visual representations

Pre-train a visual **de-tokenizer** for image generation

Two-stage training:

- Semantic reconstruction
- Fine-grained detail recovery via conditional image



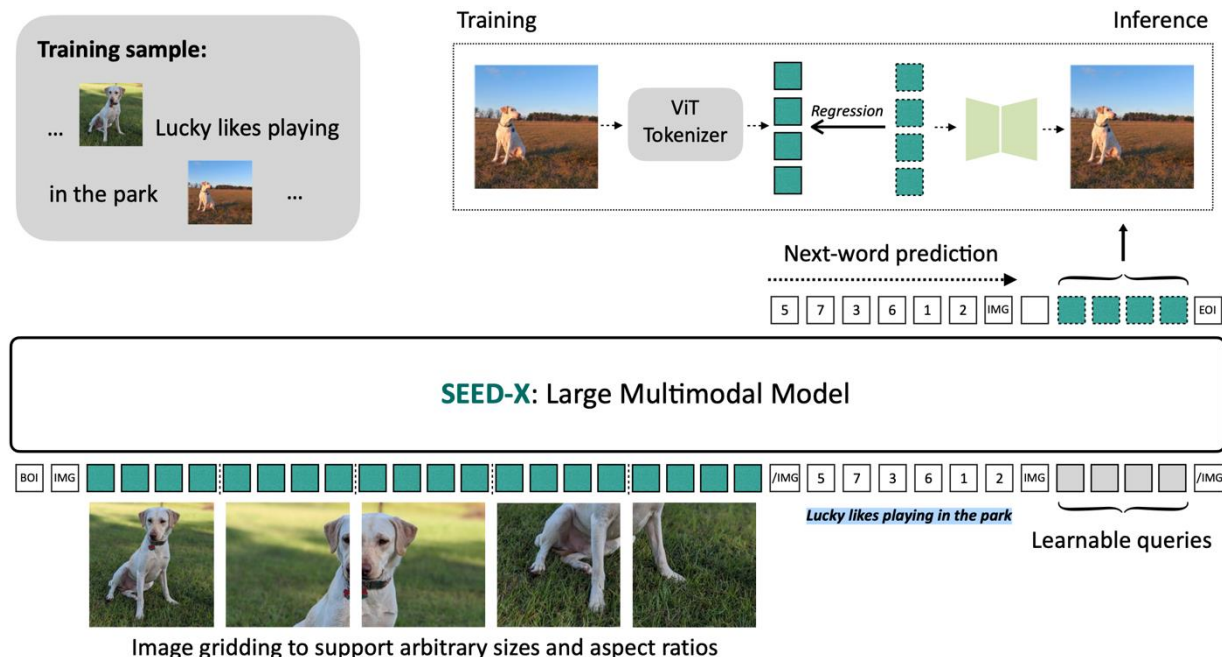
Visual Autoregressive - SEED-X (Tencent)

Input: Interleave text tokens and dynamic resolution image features

Training objectives

- *Next-word prediction (text)*
- *Visual feature regression (learnable queries $\rightarrow$ ViT features)*

Inference: LLM outputs visual features $\rightarrow$ visual de-tokenizer generates images

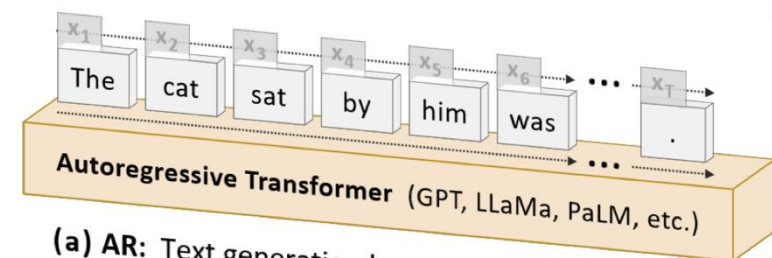


Visual Scale Autoregressive - VAR (Byte Dance)

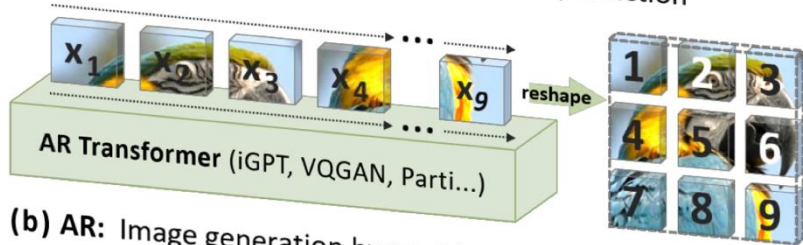
Visual Autoregressive: *Flattens 2D images into a 1D "sentence" and predicts the next tokens or words sequentially*

Visual Scale Autoregressive: *Generates entire images at different resolutions in a progressive manner*

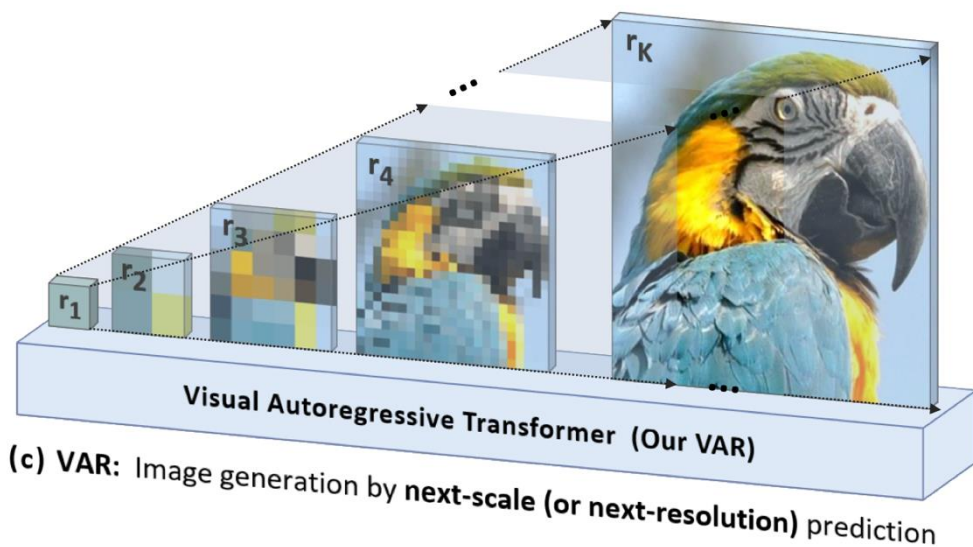
Three Different Autoregressive Generative Models



(a) AR: Text generation by **next-token** prediction



(b) AR: Image generation by **next-image-token** prediction



(c) VAR: Image generation by **next-scale (or next-resolution)** prediction

Tian K, Jiang Y, Yuan Z, et al. Visual autoregressive modeling: Scalable image generation via next-scale prediction

Visual Scale Autoregressive - VAR (Byte Dance)

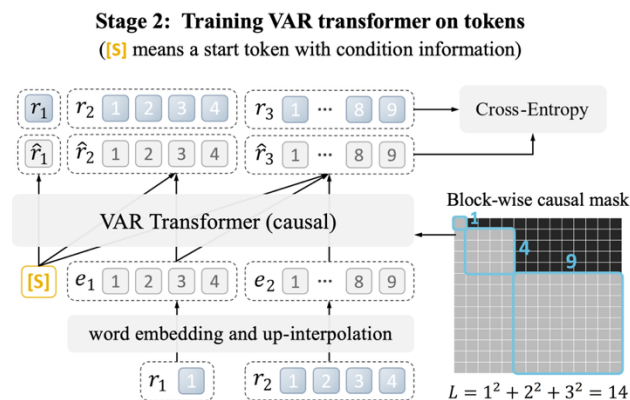
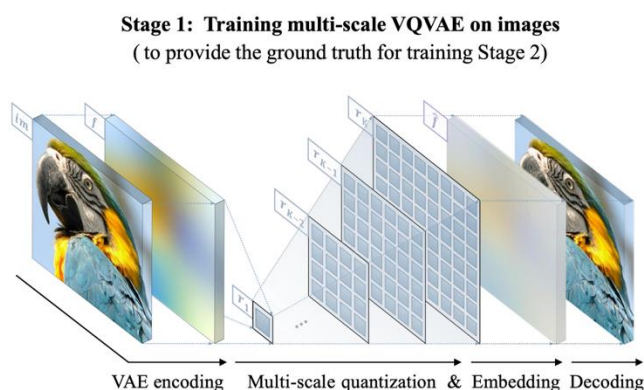
Training Paradigm

Stage 1: Multi-scale VQ-VAE

- Encode each image into a hierarchy of discrete token maps
- Each token map corresponds to a specific resolution level

Stage 2: VAR Transformer

- Autoregressively predicts images across scales
- Input: $[S], r_1, r_2, \dots, r_{k-1}$
- Output: r_k (the next-scale token map)

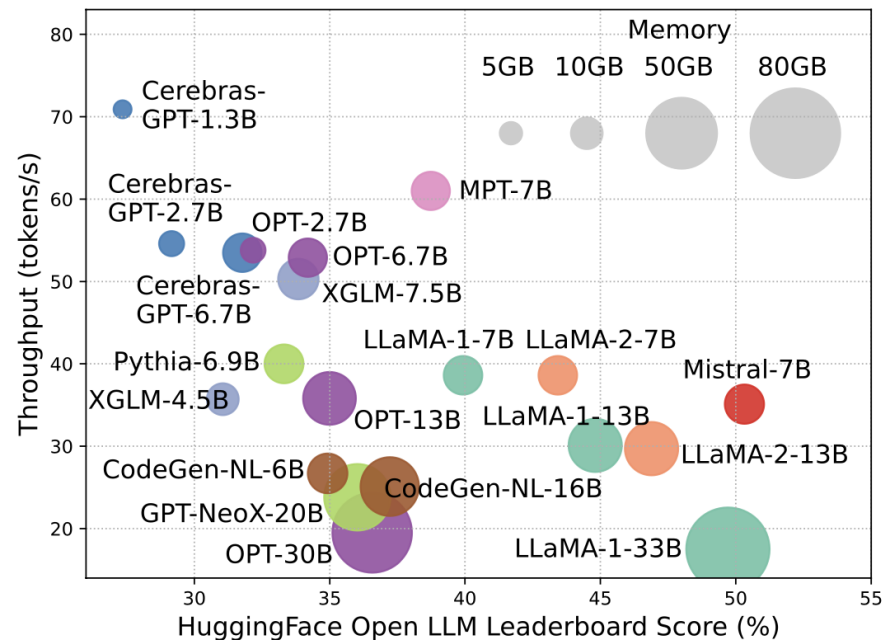
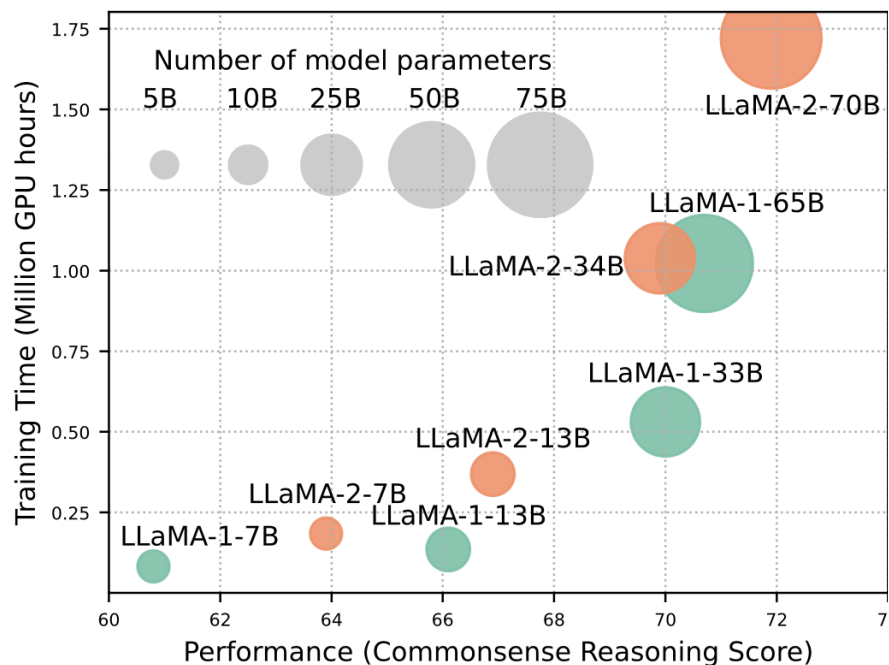


Tian K, Jiang Y, Yuan Z, et al. Visual autoregressive modeling: Scalable image generation via next-scale prediction

LLM Efficiency

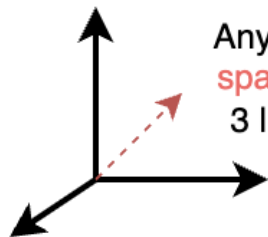
LoRA (Training), KV-Cache (Inference)

How important of efficiency ?

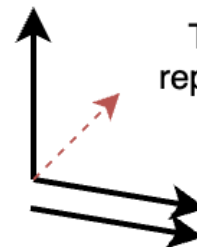


Low-Rank Adaption: LoRA

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}, \quad \text{Rank}(\mathbf{A}) = 3. \quad \mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}, \quad \text{Rank}(\mathbf{A}) = 2.$$



Any **vectors in 3-dimensional space** could be represented by 3 linear dependent vectors

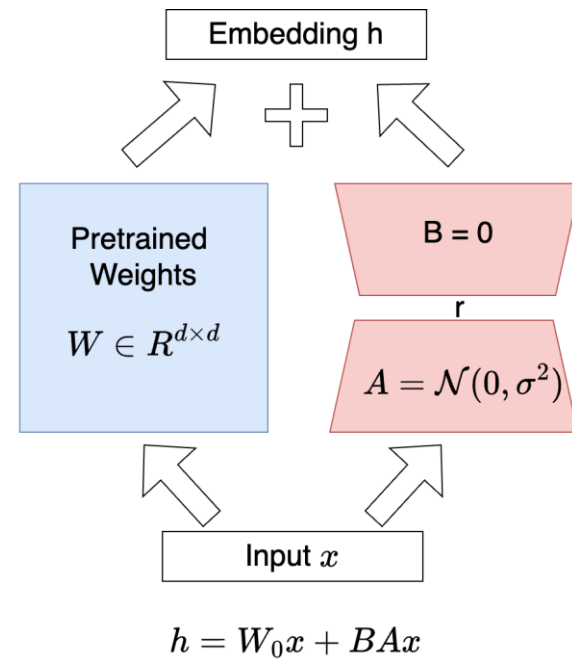
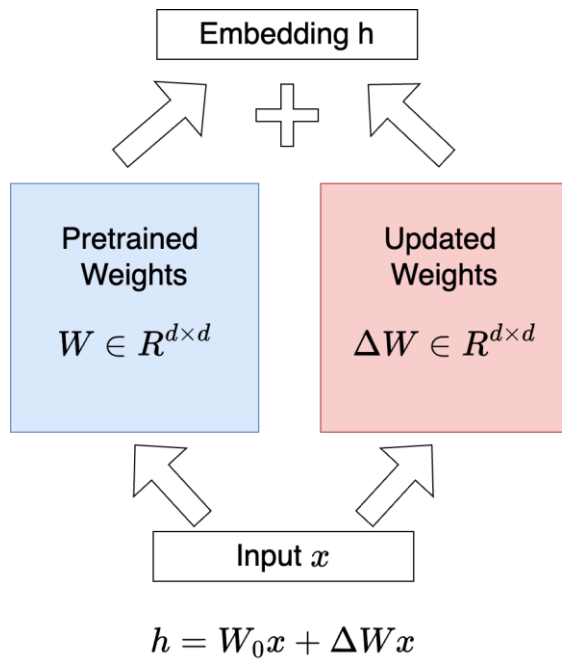


The **vector** could **not** be represented by only 2 linear dependent vectors

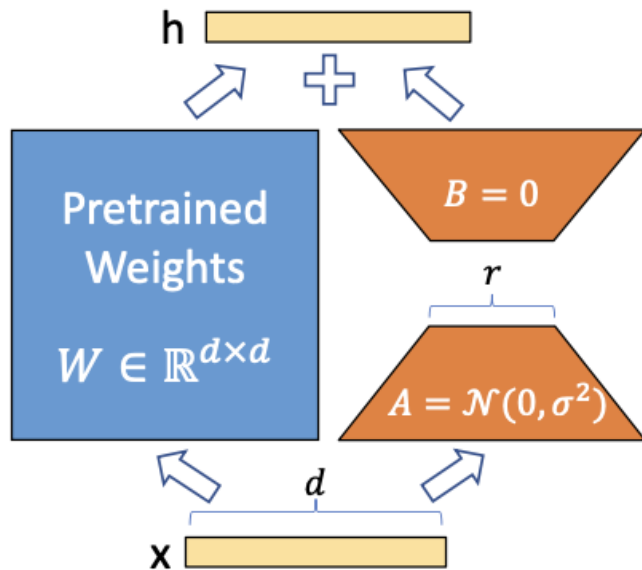
We hope the rank is large in weights as it can represent more dimensional latent features

The weights pretrained only need specific base vectors for specific tasks

Low-Rank Adaption: LoRA



Low-Rank Adaption: LoRA



$$h = W_0 x + \Delta W x = W_0 x + B A x$$

Why $B = 0$ Tuning starts from original W

Why $A = \mathcal{N}(0, \sigma^2)$ Gradient updating process is stable

Implement for each layer

1) This is a simple example module where we want add LoRA modules to each Linear layer

```
class FeedForwardNetwork(torch.nn.Module):
    def __init__(self, embedding_dim):
        super().__init__()
        self.network = nn.Sequential(
            torch.nn.Linear(embedding_dim, 4*embedding_dim),
            torch.nn.ReLU(),
            torch.nn.Linear(4*embedding_dim, embedding_dim),
            torch.nn.Dropout(0.2)
        )
    def forward(self, x):
        return self.network(x)
```

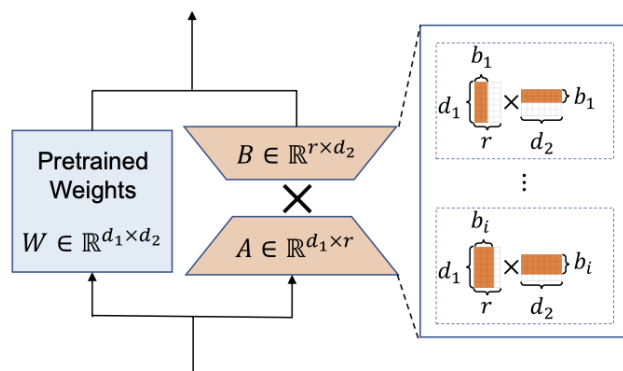
2) The original Linear layer is provided as input to the LinearWithLoRA layer

```
class LinearWithLoRA(torch.nn.Module):
    def __init__(self, linear, rank, alpha):
        super().__init__()
        self.linear = linear
        self.lora = LoRALayer(
            linear.in_features, linear.out_features, rank, alpha
        )
    def forward(self, x):
        return self.linear(x) + self.lora(x)
```

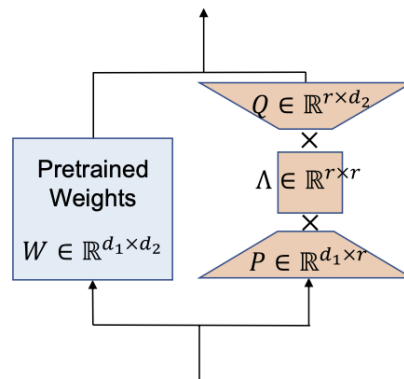
3) The LinearWithLoRA layer consists of the original Linear layer plus a LoRALayer containing the LoRA weight matrices

4) In the new LinearWithLoRA layer, we add the linear and LoRA layer outputs during the forward pass.

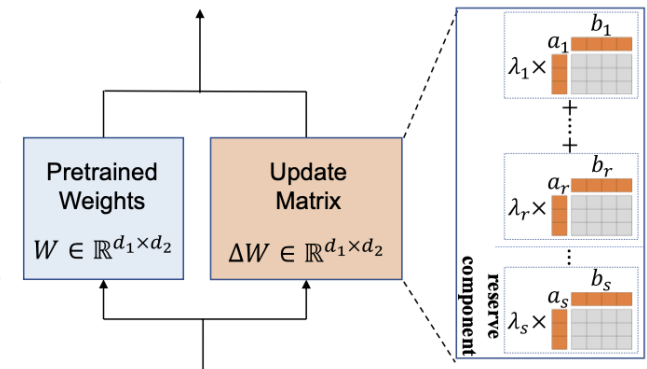
Other Variants



(a) DyLoRA

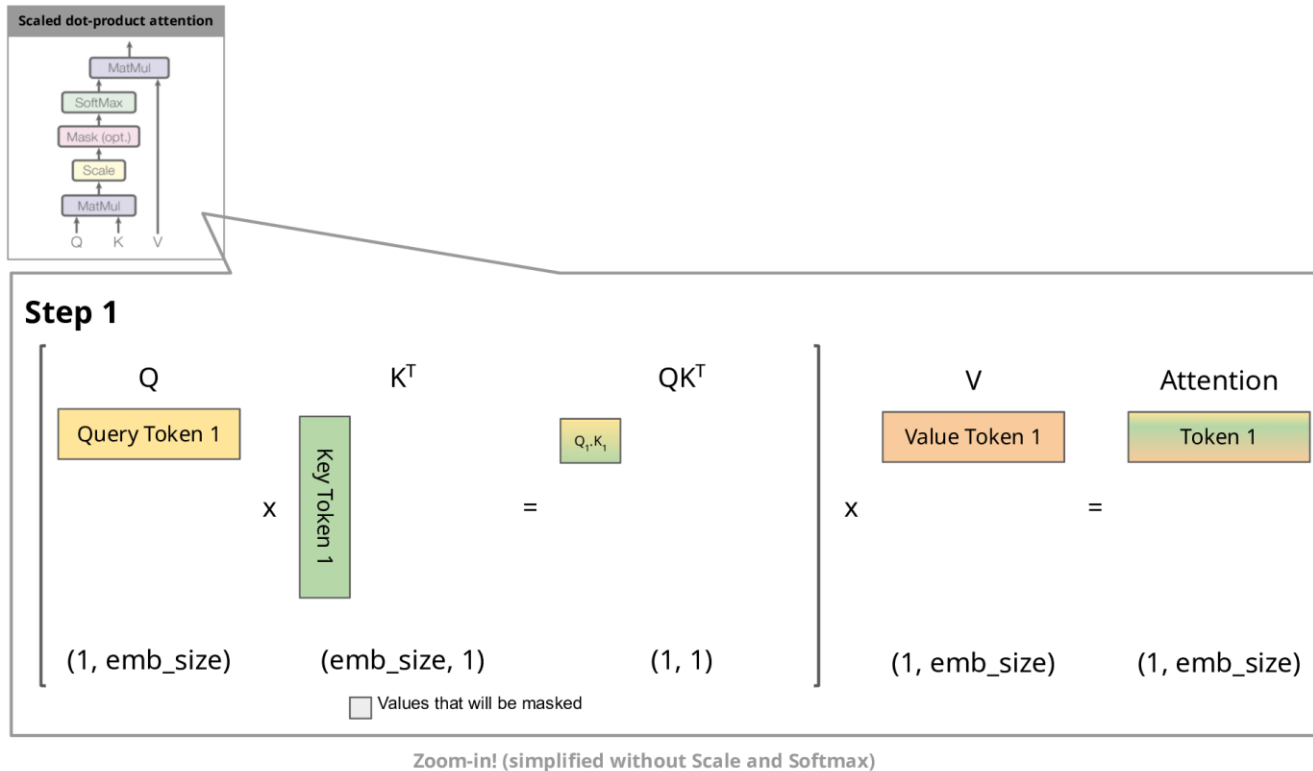


(b) AdaLoRA

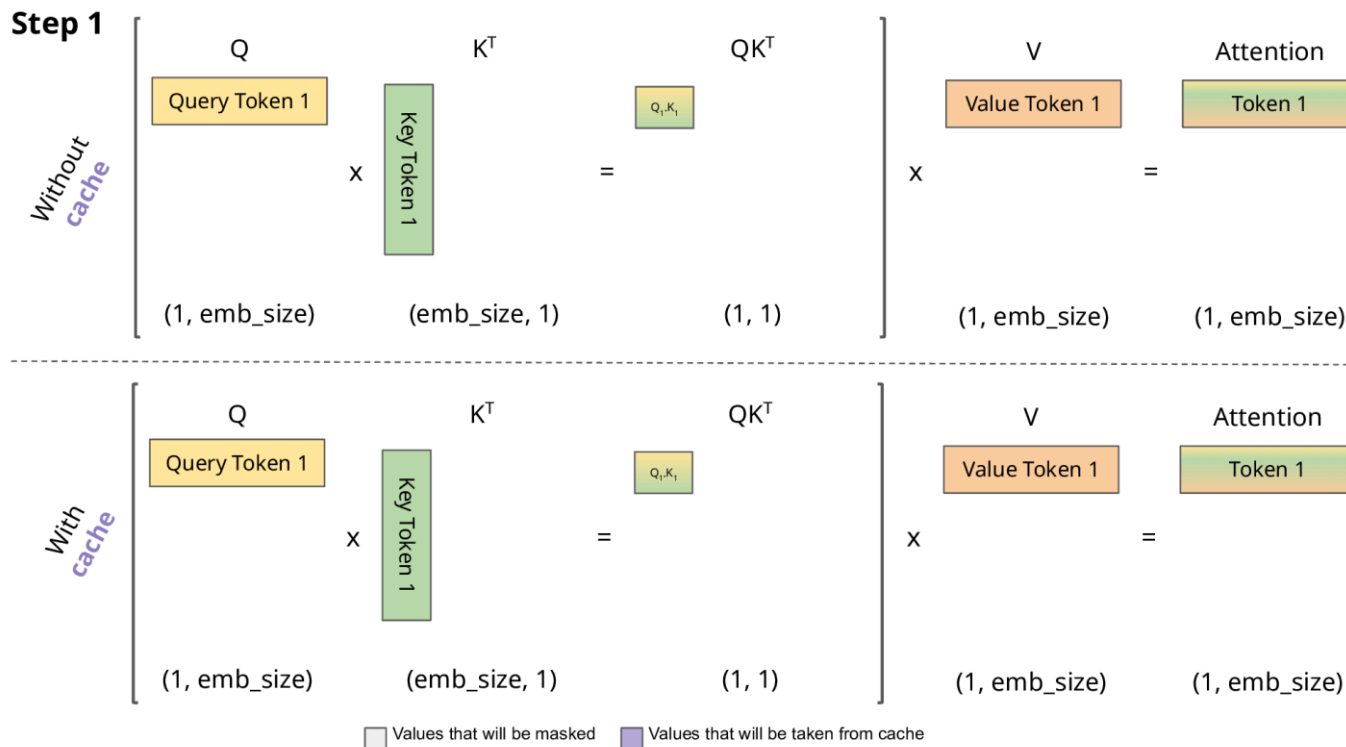


(c) IncreLoRA

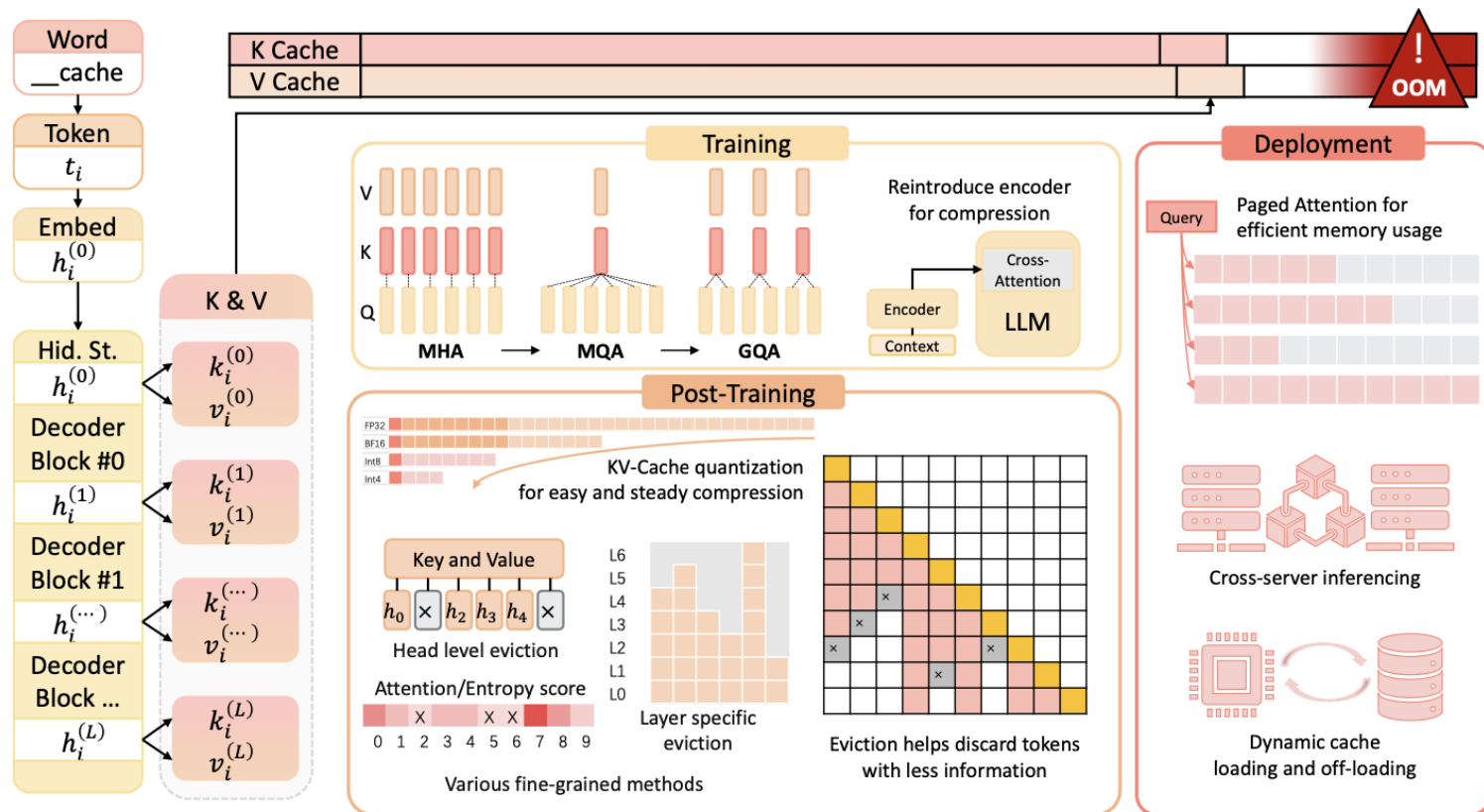
KV Cache Inference



KV Cache Inference

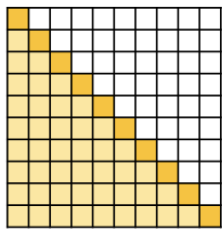


KV Cache Inference



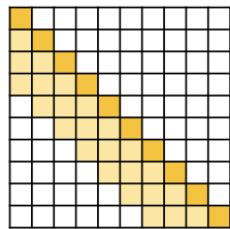
KV Cache Inference

Traditional Policies

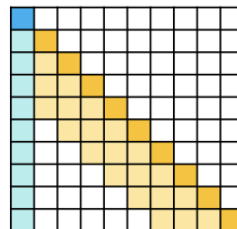


(a) Dense Attention

Static Policies

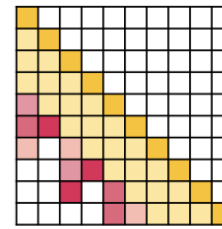


(b) Window Attention

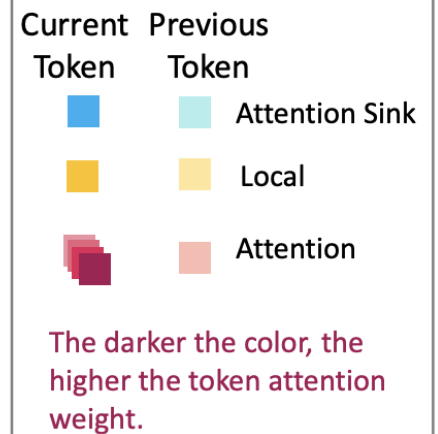


(c) Initial Token Attention

Dynamic Policies



(e) Attention Based



Mixture-of-Expert (MoE)

General Assumptions

- More parameters, greater performance

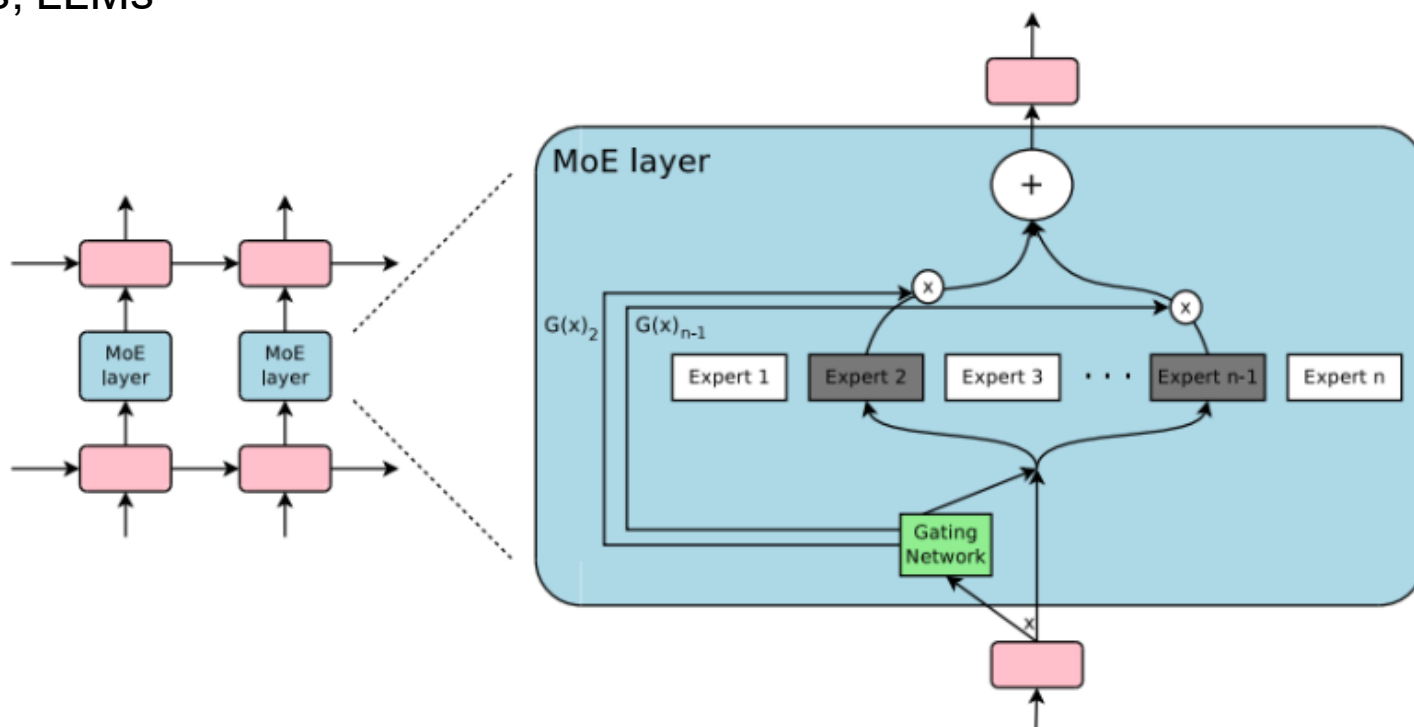
Model	Size	Code	Commonsense Reasoning	World Knowledge	Reading Comprehension	Math	MMLU	BBH	AGI Eval
MPT	7B	20.5	57.4	41.0	57.5	4.9	26.8	31.0	23.5
	30B	28.9	64.9	50.0	64.7	9.1	46.9	38.0	33.8
Falcon	7B	5.6	56.1	42.8	36.0	4.6	26.2	28.0	21.2
	40B	15.2	69.2	56.7	65.7	12.6	55.4	37.1	37.0
LLAMA 1	7B	14.1	60.8	46.2	58.5	6.95	35.1	30.3	23.9
	13B	18.9	66.1	52.6	62.3	10.9	46.9	37.0	33.9
	33B	26.0	70.0	58.4	67.6	21.4	57.8	39.8	41.7
	65B	30.7	70.7	60.5	68.6	30.8	63.4	43.5	47.6
LLAMA 2	7B	16.8	63.9	48.9	61.3	14.6	45.3	32.6	29.3
	13B	24.5	66.9	55.4	65.8	28.7	54.8	39.4	39.1
	34B	27.8	69.9	58.7	68.0	24.2	62.6	44.1	43.4
	70B	37.5	71.9	63.6	69.4	35.2	68.9	51.2	54.2

- More parameters, larger inference time

Mixture of Experts

What we have:

1. Large number of parameters
2. Low inference time
3. Good performance
4. Applicable to Diffusion Models, LLMs



MoE details - Sparsity

1. We add some noise

$$H(x)_i = (x \cdot W_g)_i + \text{StandardNormal}() \cdot \text{Softplus}((x \cdot W_{\text{noise}})_i)$$

2. We only pick the top k

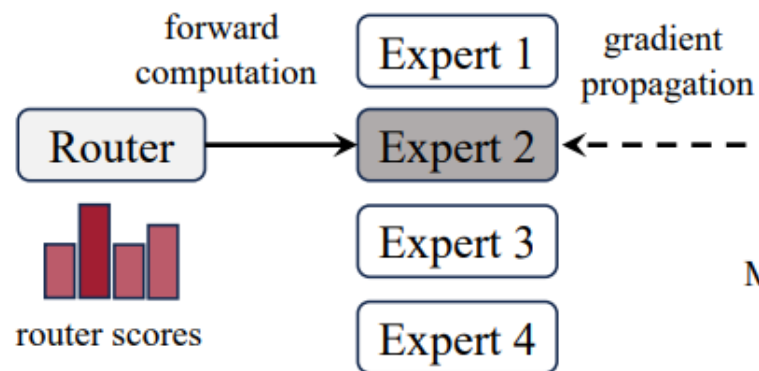
$$\text{KeepTopK}(v, k)_i = \begin{cases} v_i & \text{if } v_i \text{ is in the top } k \text{ elements of } v, \\ -\infty & \text{otherwise.} \end{cases}$$

3. We apply the softmax.

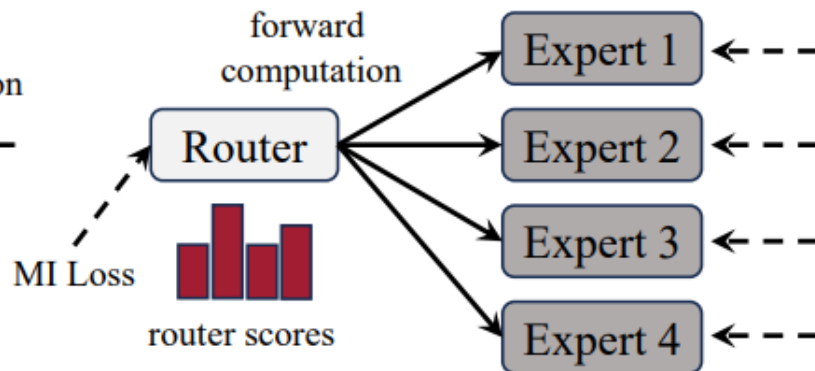
$$G(x) = \text{Softmax}(\text{KeepTopK}(H(x), k))$$

MoE details – Recent approaches

DS-MoE: Dense training, but sparse inference



(a) Sparse Training

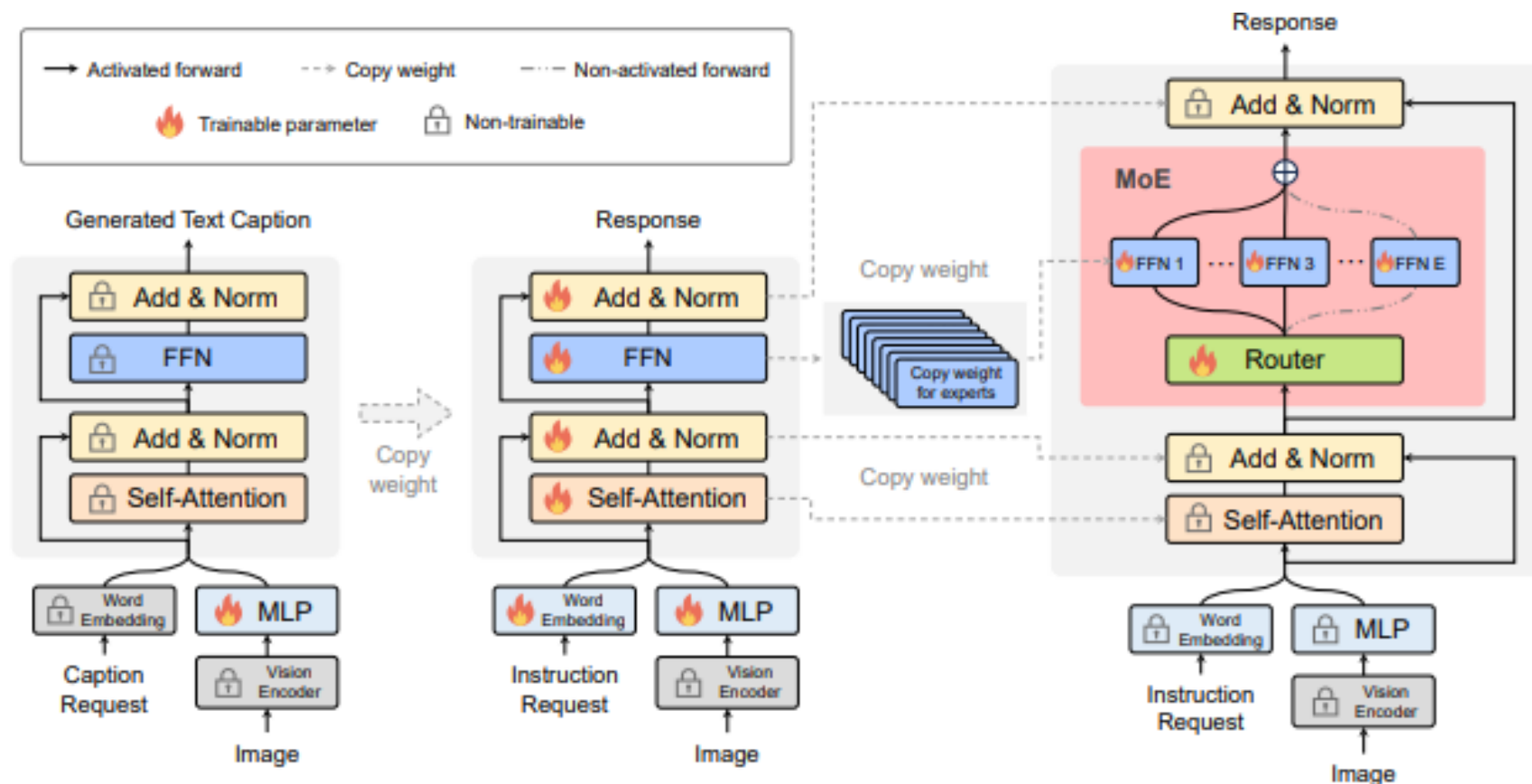


(b) Dense Training

Pan, Bowen, et al. "Dense Training, Sparse Inference: Rethinking Training of Mixture-of-Experts Language Models." *arXiv preprint arXiv:2404.05567* (2024).

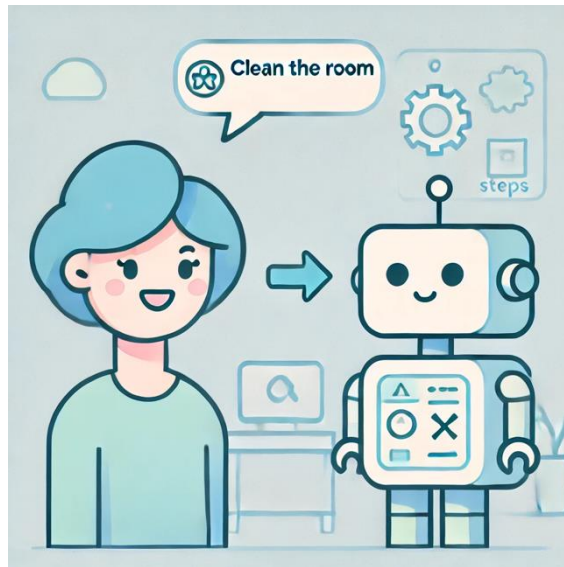
MoE details – Recent approaches

MoE-Llava: Mixture of Experts for Large Vision-Language Models



Lin, Bin, et al. "Moe-llava: Mixture of experts for large vision-language models." *arXiv preprint arXiv:2401.15947* (2024).

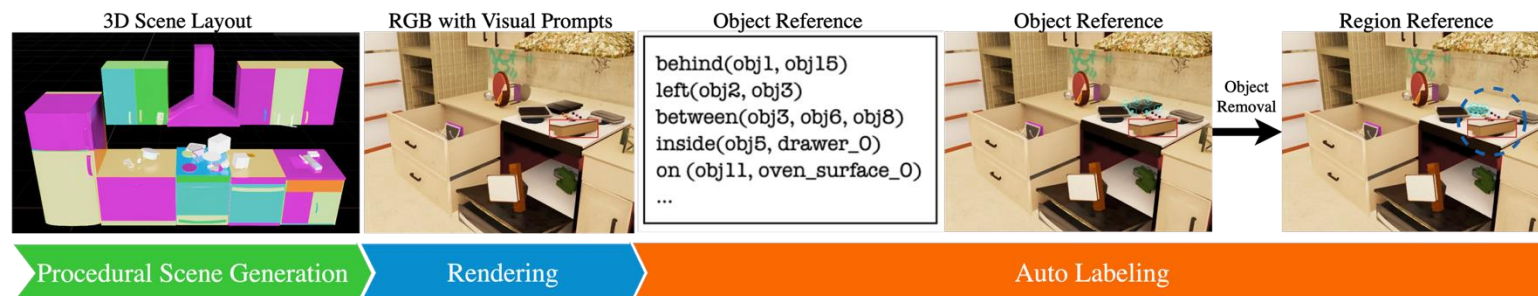
Multi-modal Foundation Model in Robotics



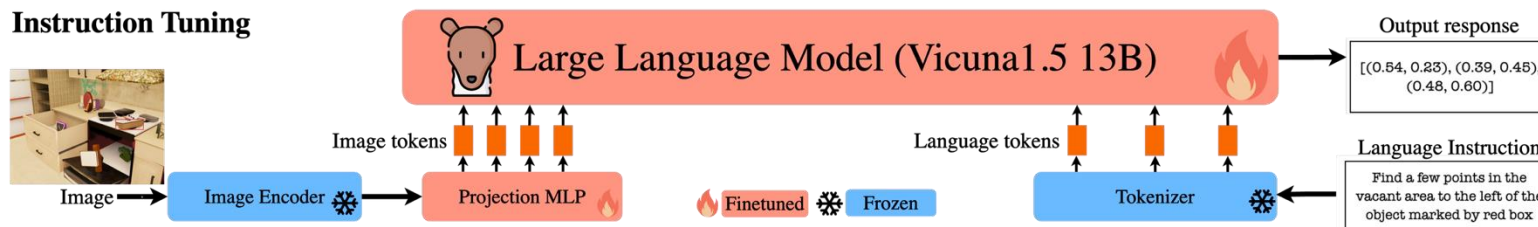
Source: DALL-E

RoboPoint: VLM for Action Points

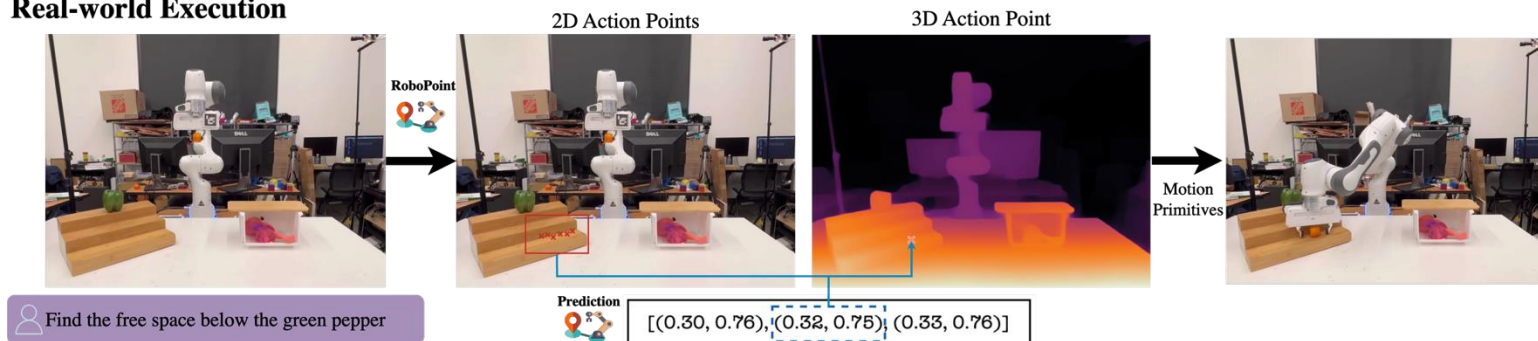
Data Generation



Instruction Tuning

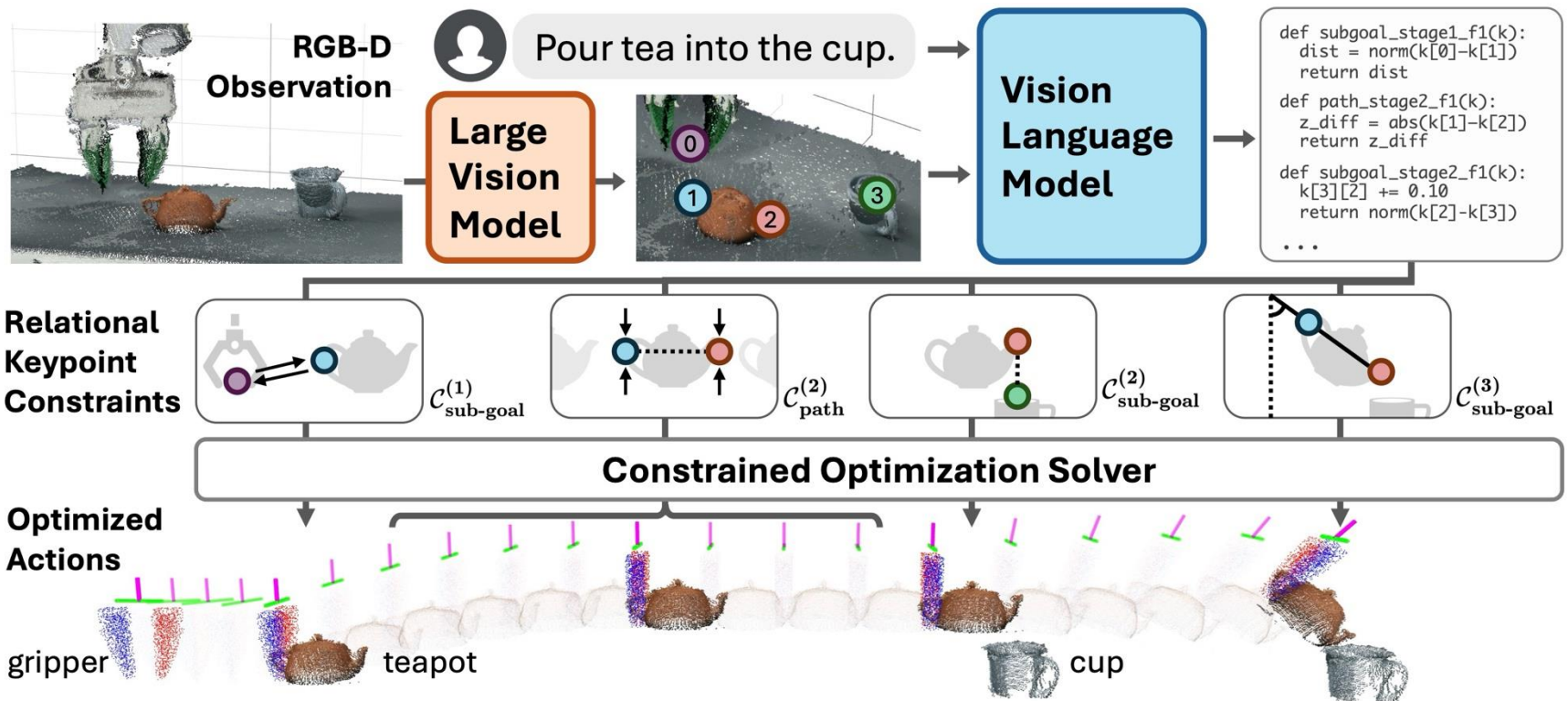


Real-world Execution



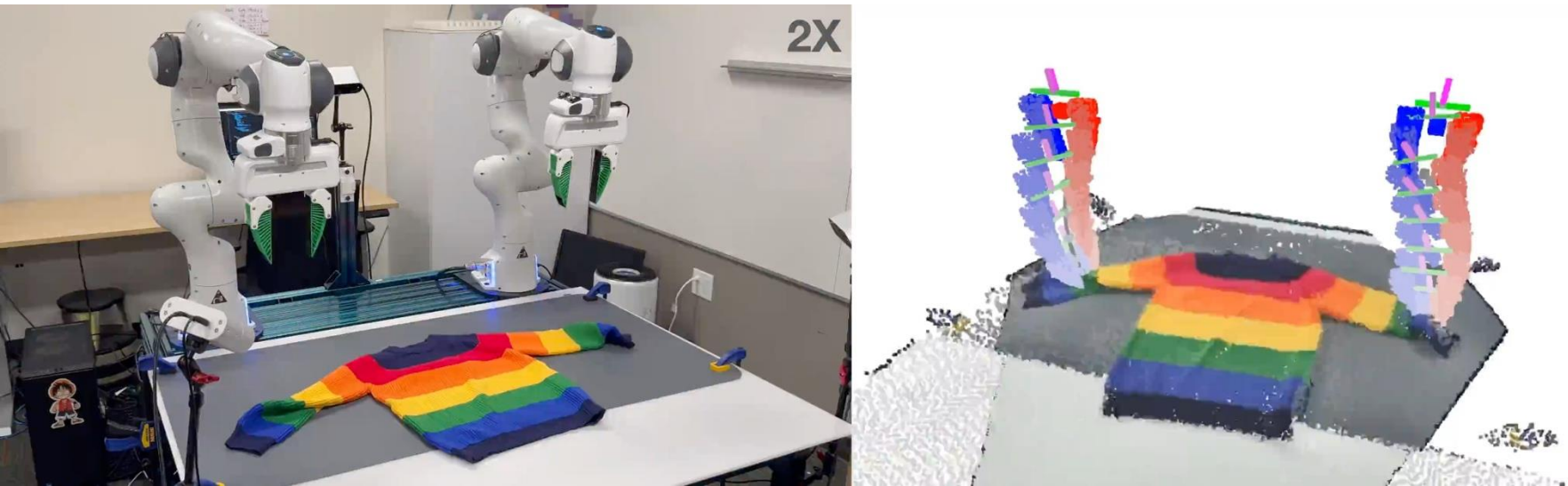
RoboPoint: A Vision-Language Model for Spatial Affordance Prediction for Robotics. CoRL 2024.

ReKep: VLM for Keypoint Constraints



ReKep: Spatio-Temporal Reasoning of Relational Keypoint Constraints for Robotic Manipulation. CoRL 2024.

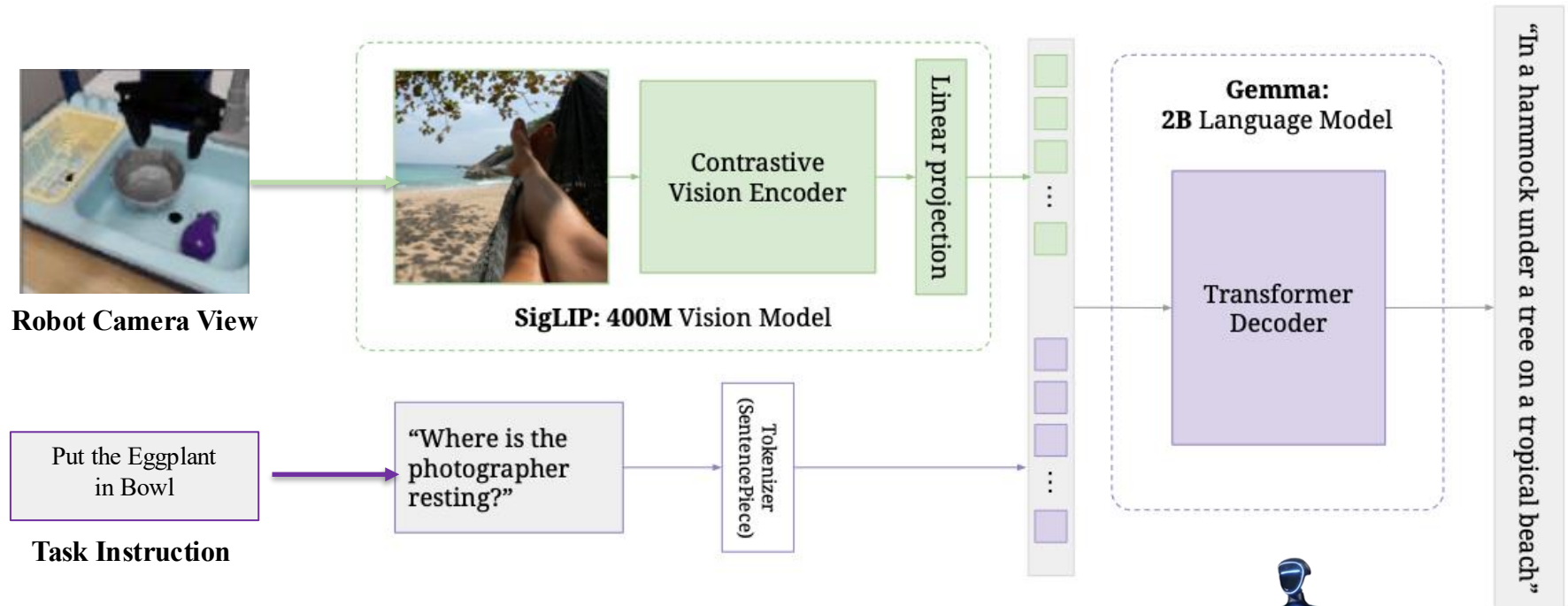
ReKep: VLM for Keypoint Constraints



ReKep: Spatio-Temporal Reasoning of Relational Keypoint Constraints for Robotic Manipulation. Sep 2024. CoRL.

From VLM to VLA

- PaliGemma (3B) by Google

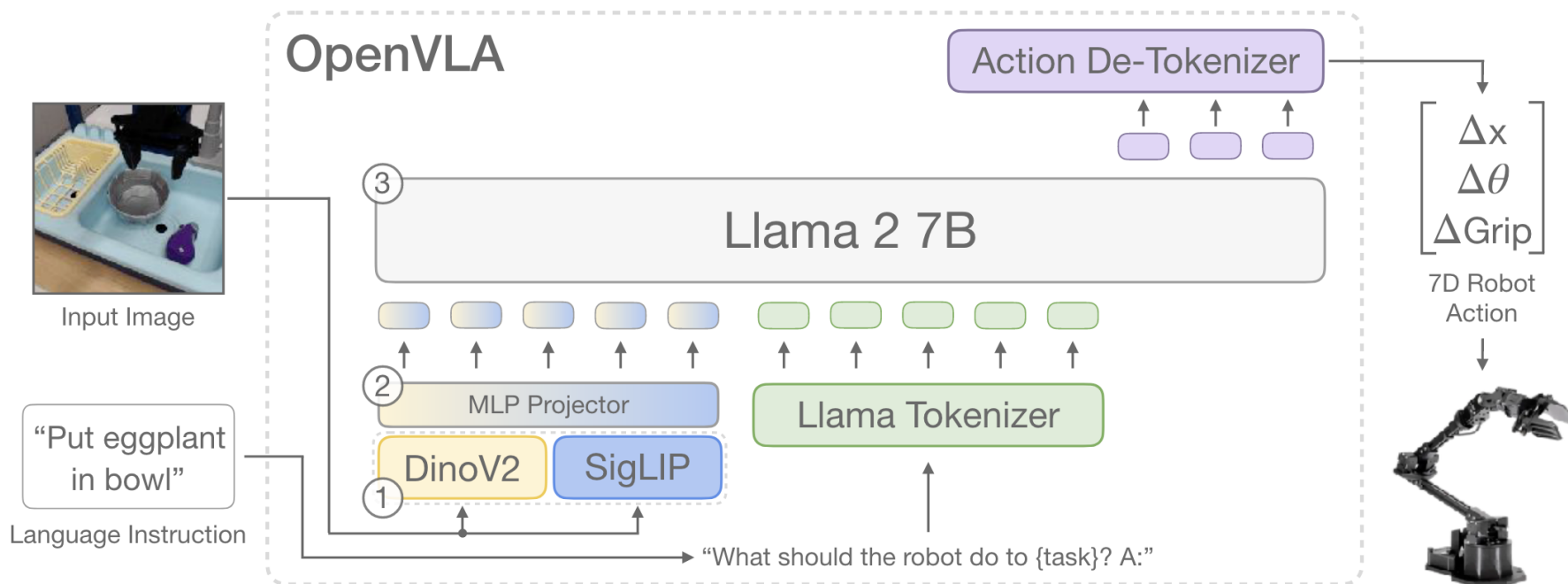


<https://deepmind.google/models/gemma/paligemma-2/>



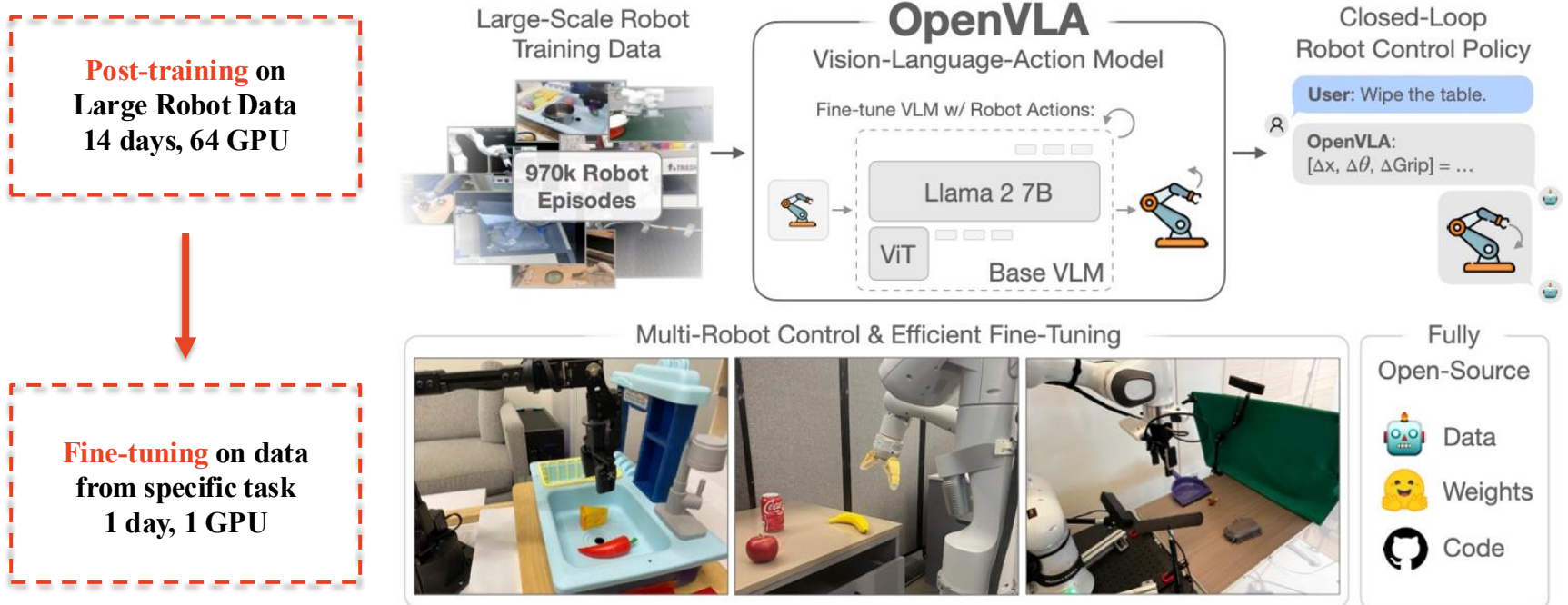
VLA Architecture

- VLA interpret both visual data and language instructions to perform actions

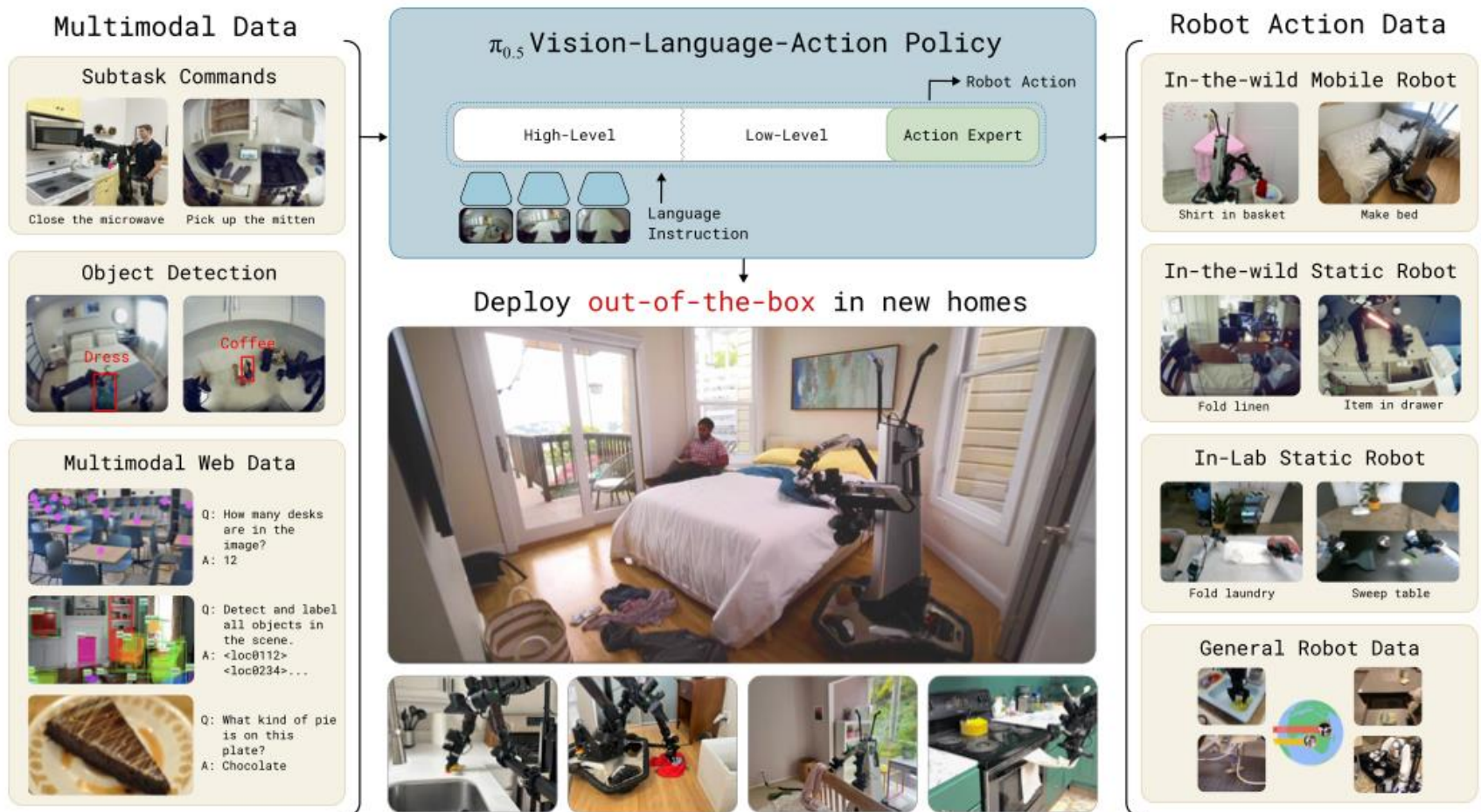


OpenVLA: An Open-Source Vision-Language-Action Model, CoRL, 2024, Stanford & Google.

VLA Training



$\pi_{0.5}$: VLA with Open-World Generalization



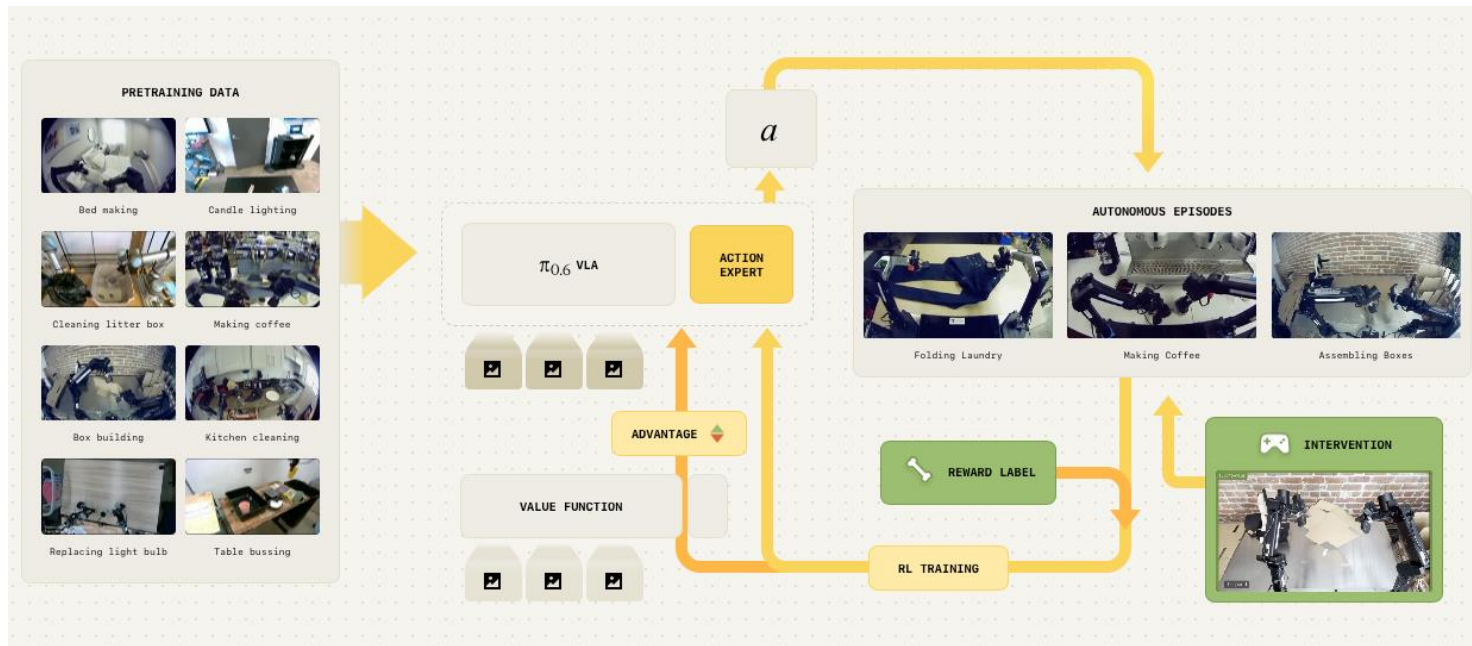
$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization. Apr 2025. Physical Intelligence.

$\pi_{0.5}$: VLA with Open-World Generalization



$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization. Apr 2025. Physical Intelligence.

$\pi_{0.6}$: a VLA That Learns from Experience



Thank you!